

META Local No-Jump Diffusion Analysis

Ticker: META

Date range: 2021-06-30 09:31:00 to 2025-12-31 15:59:00

Frequencies analyzed in the current output: daily only

Reported sample split uses an RSD-filtered daily drift definition: detectable_upward_drift, detectable_downward_drift

Daily local estimation window: past 63 segment log-VWAP increments (1/3-day observations)

Daily future horizons: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] days

Daily detail horizons: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13] half-hour units over the first trading day

Daily quantile lookback: 60 trading days

Abnormal-volume threshold: rolling 99th percentile throughout the report.

Location groups used in the figures: only above-mid abnormal, only below-mid abnormal, and a general baseline

At-mid-only cases are excluded from the main comparison figures.

Future-sequence return definition: simple cumulative return from signal-time VWAP to horizon-end VWAP, i.e.

Cumulative return at time t means total return from signal time up to t, not the one-step return for only t

This report does not run intraday minute-level post-signal analysis.

Daily future returns use aggregated daily VWAP built from minute VWAP.

Daily segmentation: 09:30-11:30, 11:30-13:30, 13:30-16:00 ET

Daily drift-strength test uses 1-minute price-difference returns $r_{\{d,i\}} = p_{\{d,i\}} - p_{\{d,i-1\}}$.

$RSD_d(k) = (1 / \Delta) * \sum r_{\{d,i\}} r_{\{d,i-k\}}$; $RQ_d = (1 / (3 * \Delta)) * \sum r_{\{d,i\}}^4$.

$VarHat_d = RQ_d / \Delta$ and $ST_d(k) = RSD_d(k) / \sqrt{RQ_d / \Delta}$.

Daily detectable-drift screen: $abs(ST_d(k)) > 1.96$, with lag $k = 1$.

The sign of μ_hat_d is used only after the daily RSD screen passes.

Method note: the local process is treated as a no-jump diffusion on log VWAP with discrete observations.

Estimation note: the likelihood uses the first-order Ait-Sahalia approximate transition density on log-VWAP

Under the local constant-parameter specification this reduces to a Gaussian increment likelihood with close

Observation note: all prices are VWAP-based; segment VWAP uses volume-weighted minute VWAP when volume exists

Color map used throughout: red = only above-mid abnormal, blue = only below-mid abnormal, black = general b

Data directories: count=data_2020_2025_count, main=data_2020_2025

META Executive Summary

Daily drift is now defined by an RSD-based daily drift-strength screen, not by sign alone.

A day is classified as detectable upward/downward drift only when $\text{abs}(\text{ST}_d(k)) > 1.96$; otherwise it enters

The main comparison in each drift-sign section is: only above-mid abnormal volume vs only below-mid abnormal

Daily drift-state counts: $\text{no_detectable_drift}=525$, $\text{detectable_upward_drift}=339$, $\text{detectable_downward_drift}=1$

Future cumulative return uses VWAP consistently, and the current output reports only daily horizons.

Future diffusion paths are shown as cumulative sequence volatility percentages derived from sigma_hat over

Daily case counts:

Detectable Downward Drift | above mid only | $n=1$

Detectable Downward Drift | general baseline with no abnormal volume | $n=163$

Detectable Upward Drift | above mid only | $n=1$

Detectable Upward Drift | below mid only | $n=3$

Detectable Upward Drift | general baseline with no abnormal volume | $n=311$

Noise Dominant | above mid only | $n=2$

Noise Dominant | below mid only | $n=3$

Noise Dominant | general baseline with no abnormal volume | $n=470$

Lowest future-parameter p-value in the above/below/baseline comparison is 0.0113 for $\text{future_sigma_hat_8d}$ in

Triggered-day regression is active; best p-value is 0.214 for μ_hat_d on $\text{next_day_close_cum_return_d1}$.

Representative fitted-vs-realized examples are selected by lowest out-of-sample path RMSE, not by manual ch

The local benchmark is a no-jump diffusion on log VWAP, estimated from discrete log-VWAP increments via the

All observations in this module are VWAP-based; daily analysis uses three fixed market segments per day.

META Daily Drift Screen

Daily drift-strength screen summary

Total completed trading days classified: 1042

Detectable trigger threshold: $\text{abs}(\text{ST}_d(k)) > 1.96$

Drift-label counts:

no_detectable_drift: 525

detectable_upward_drift: 339

detectable_downward_drift: 178

Most common invalid reasons:

missing_timestamps;nan_or_inf_returns: 61

nan_or_inf_returns: 22

missing_timestamps: 5

insufficient_minute_bars;missing_timestamps;nan_or_inf_returns: 4

insufficient_minute_bars;nan_or_inf_returns: 3

Triggered detectable-drift days: 568

Triggered-day regression status:

$\mu_{\hat{d}}$ on next_day_close_cum_return_d1: $n=164$, slope=-0.8367, $p=0.214$, $r^2=0.0095$

$\mu_{\hat{d}}$ on future_mu_hat_1d: $n=164$, slope=0.06777, $p=0.93$, $r^2=4.83e-05$

$\mu_{\hat{d}}$ on next_day_close_cum_return_d1: $n=315$, slope=-0.06552, $p=0.303$, $r^2=0.00339$

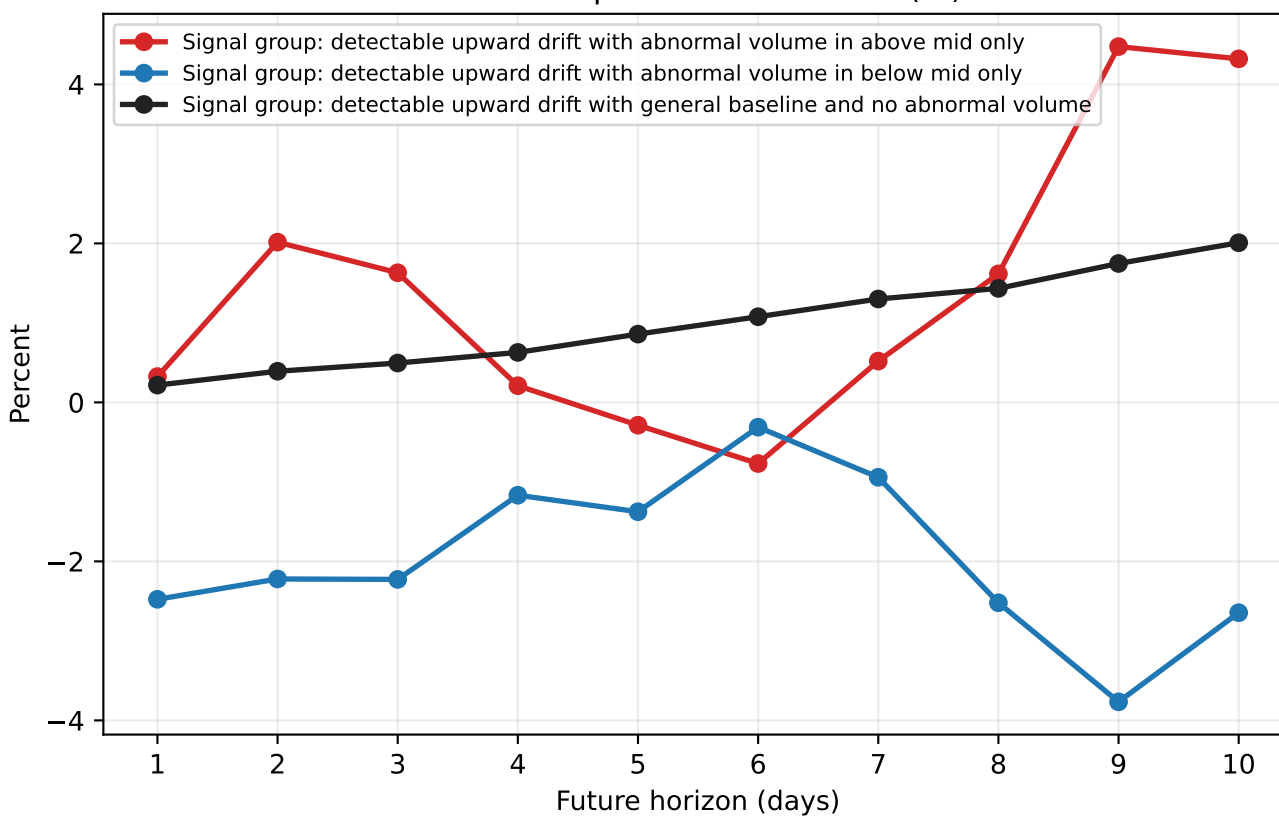
$\mu_{\hat{d}}$ on future_mu_hat_1d: $n=314$, slope=-0.03947, $p=0.516$, $r^2=0.00135$

$\mu_{\hat{d}}$ on next_day_close_cum_return_d1: $n=474$, slope=-2.186, $p=0.341$, $r^2=0.00192$

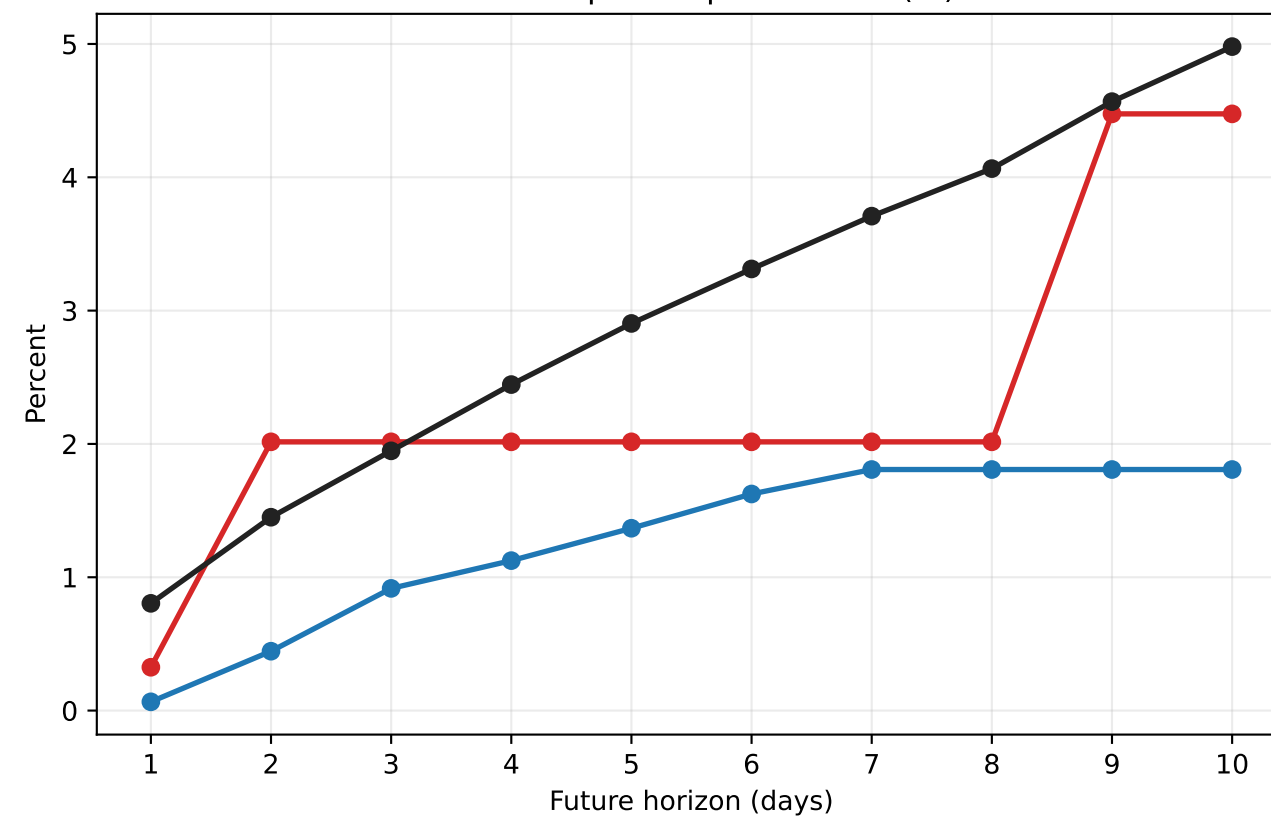
$\mu_{\hat{d}}$ on future_mu_hat_1d: $n=474$, slope=-0.5529, $p=0.239$, $r^2=0.00294$

META | Daily | Detectable Upward Drift | Future seq return and volatility by abnormal location

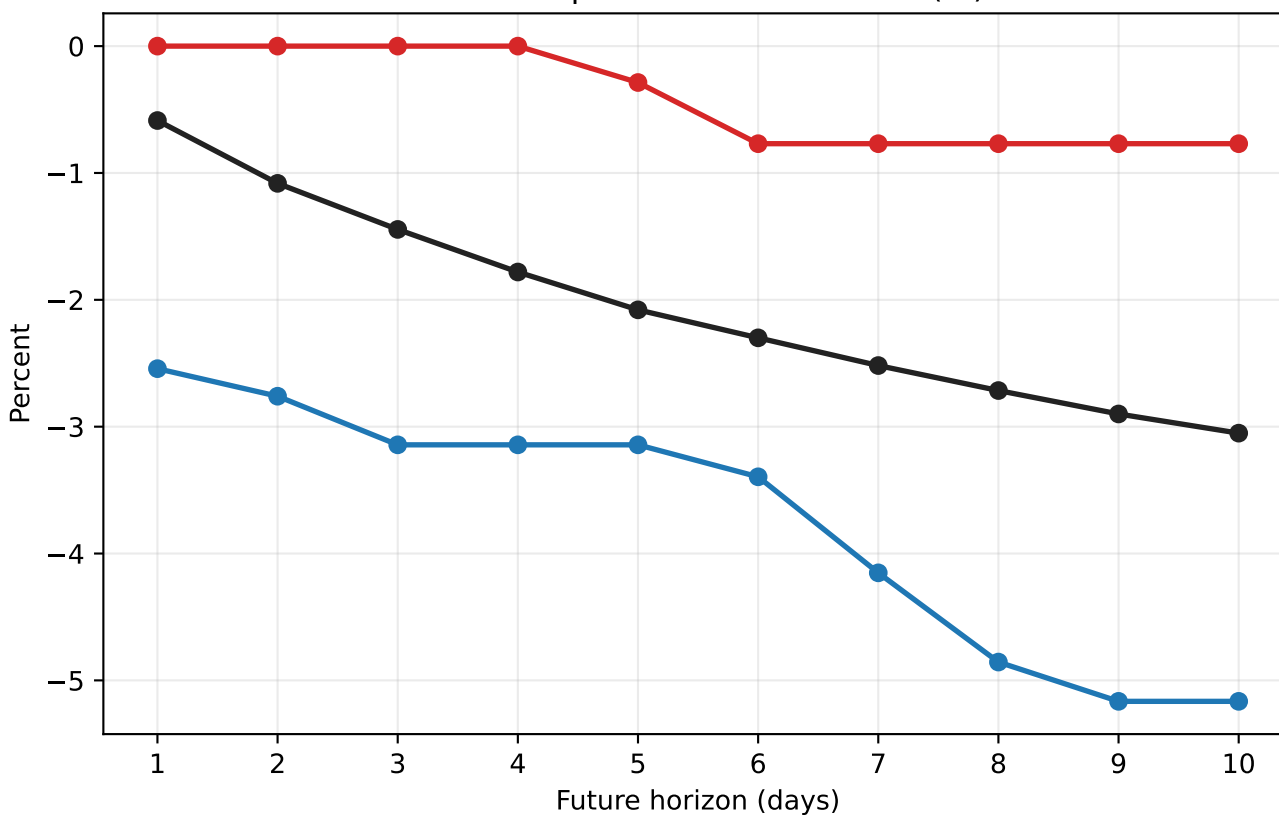
Future seq cumulative return (%)



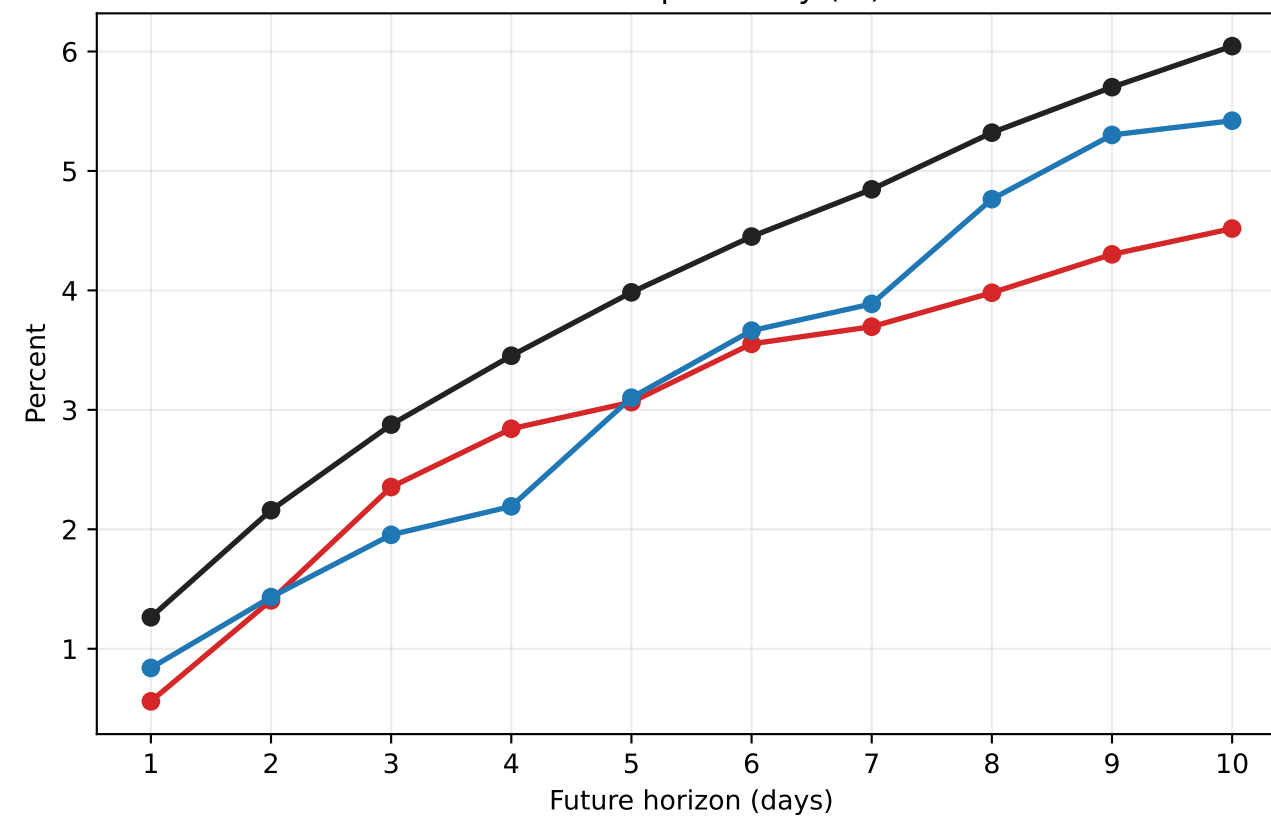
Future seq max up excursion (%)



Future seq max down excursion (%)



Future seq volatility (%)



META Daily Detectable Upward Drift Notes

Daily comparison summary

Reference group = largest-n group at terminal horizon

Reference: Signal group: detectable upward drift with general baseline and no abnormal volume

Signal group: detectable upward drift with general baseline and no abnormal volume

n=311 | pre_mu=0.008237 | pre_sigma_seq=12.41%

future_mean_return=2.01% | future_sigma_seq=6.05%

p(mu)=nan | p(sigma)=nan | p(future sigma)=nan

Signal group: detectable upward drift with abnormal volume in below mid only

n=3 | pre_mu=0.003783 | pre_sigma_seq=3.64%

future_mean_return=-2.65% | future_sigma_seq=5.42%

p(mu)=0.0394 | p(sigma)=3.71e-06 | p(future sigma)=0.292

Signal group: detectable upward drift with abnormal volume in above mid only

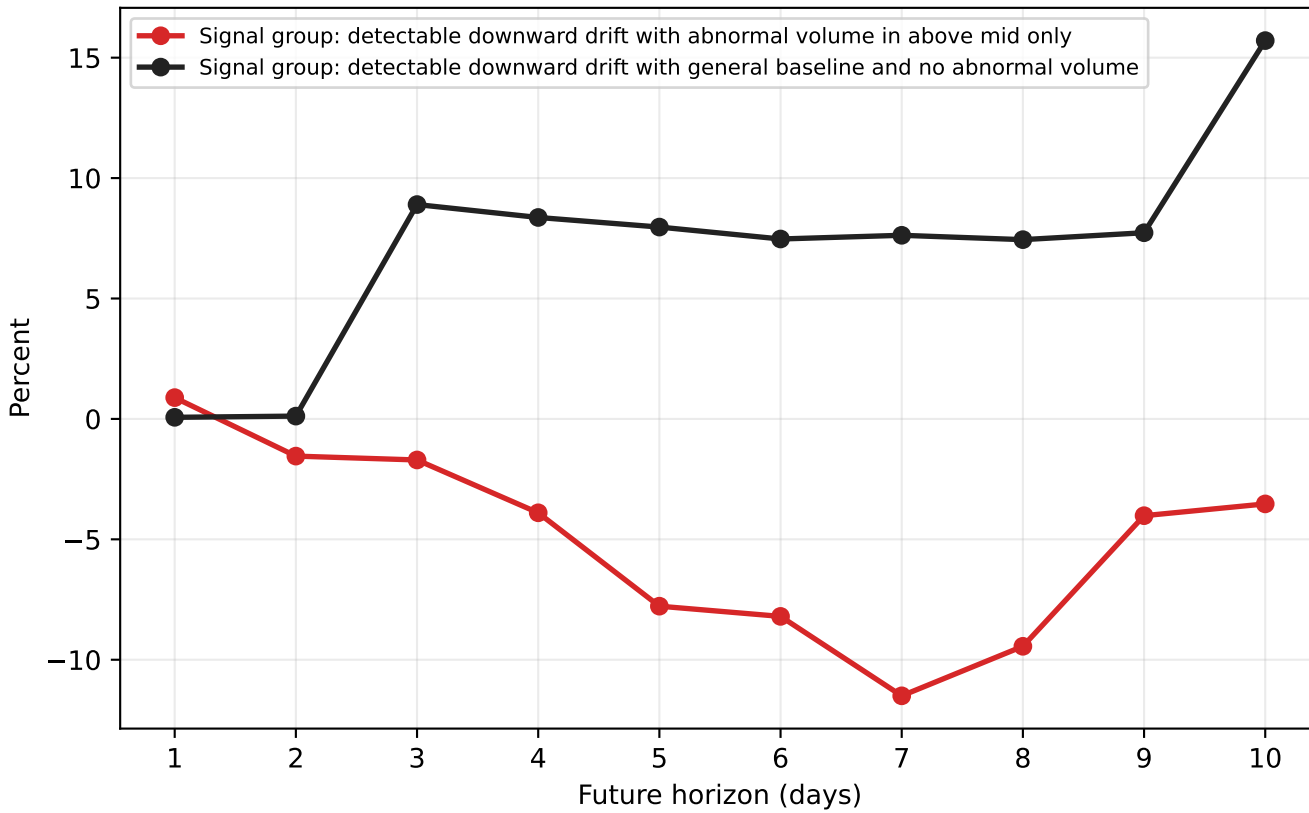
n=1 | pre_mu=0.002321 | pre_sigma_seq=4.98%

future_mean_return=4.32% | future_sigma_seq=4.52%

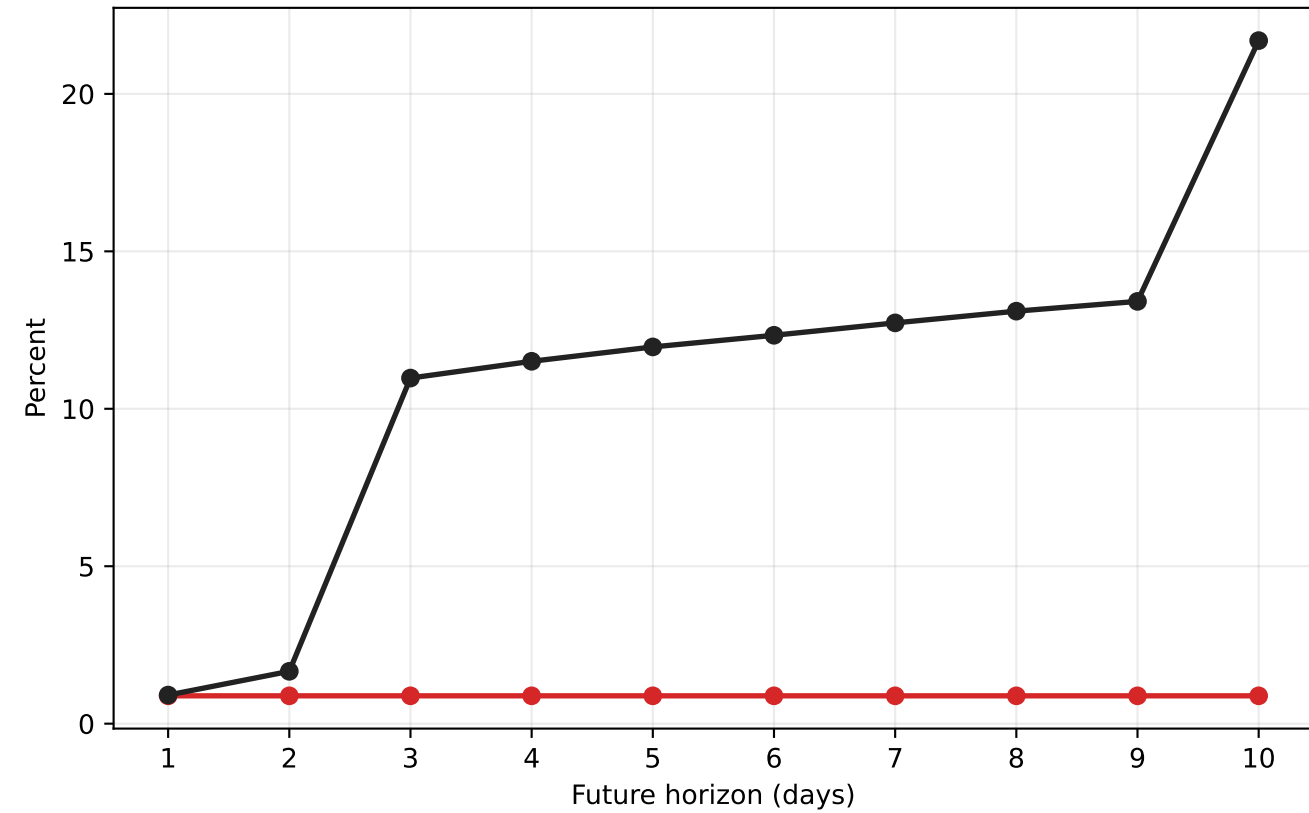
p(mu)=nan | p(sigma)=nan | p(future sigma)=nan

META | Daily | Detectable Downward Drift | Future seq return and volatility by abnormal location

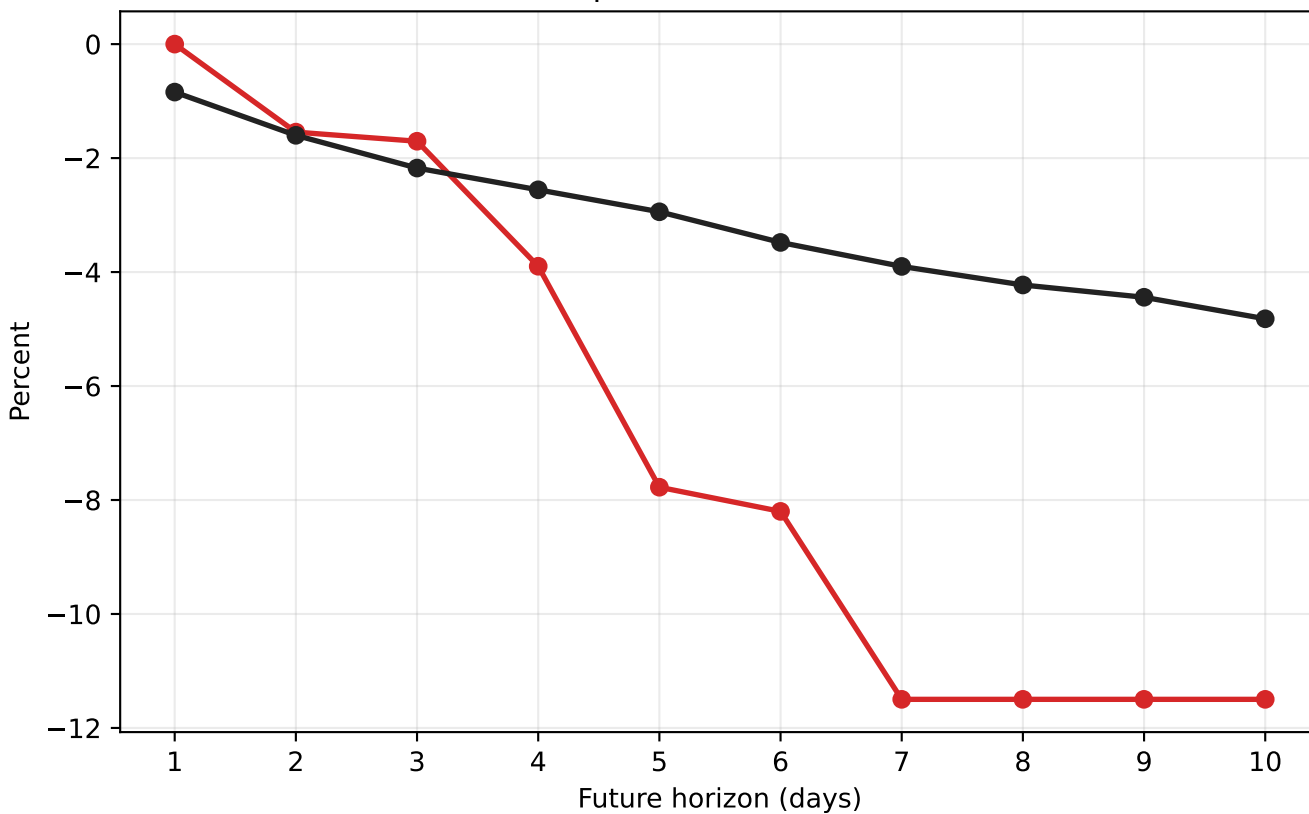
Future seq cumulative return (%)



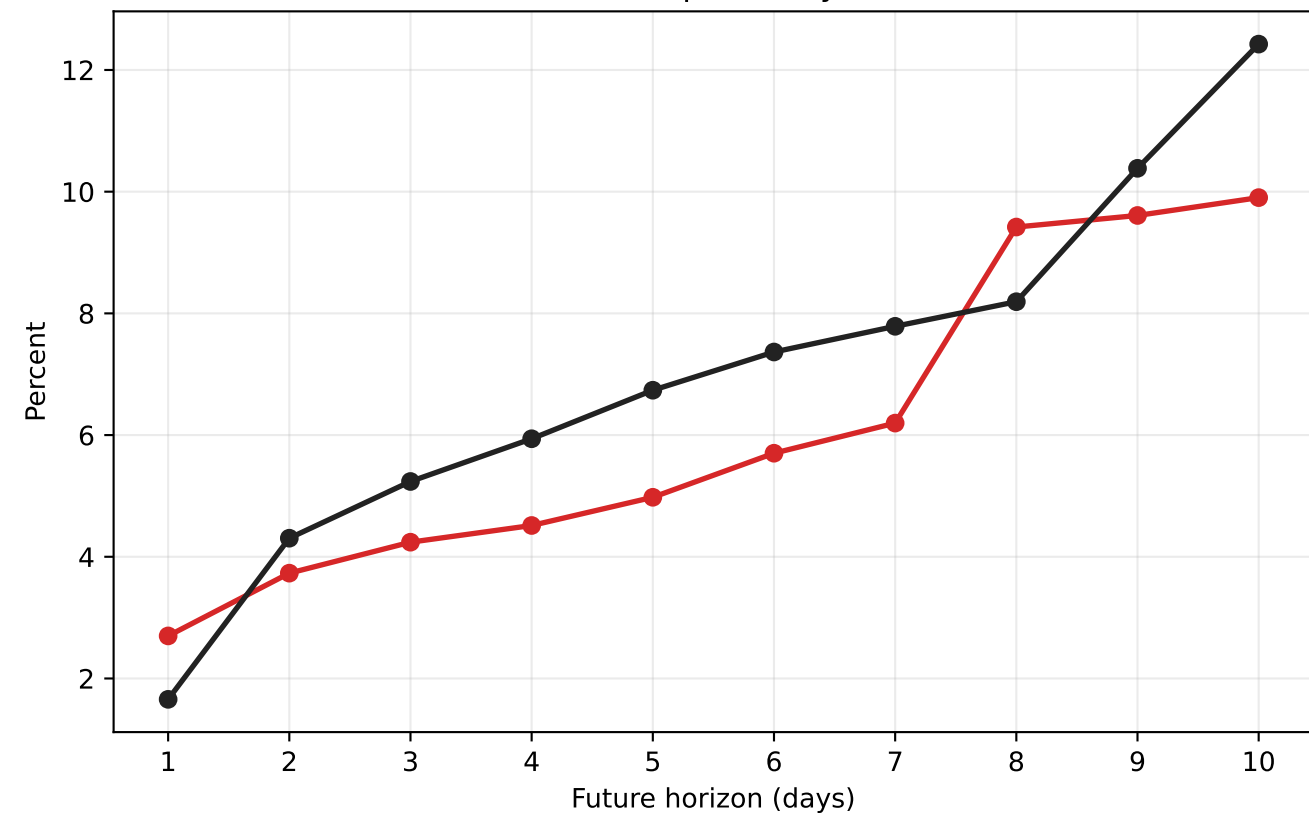
Future seq max up excursion (%)



Future seq max down excursion (%)



Future seq volatility (%)



META Daily Detectable Downward Drift Notes

Daily comparison summary

Reference group = largest-n group at terminal horizon

Reference: Signal group: detectable downward drift with general baseline and no abnormal volume

Signal group: detectable downward drift with general baseline and no abnormal volume

n=163 | pre_mu=-0.003916 | pre_sigma_seq=7.50%

future_mean_return=15.71% | future_sigma_seq=12.42%

p(mu)=nan | p(sigma)=nan | p(future sigma)=nan

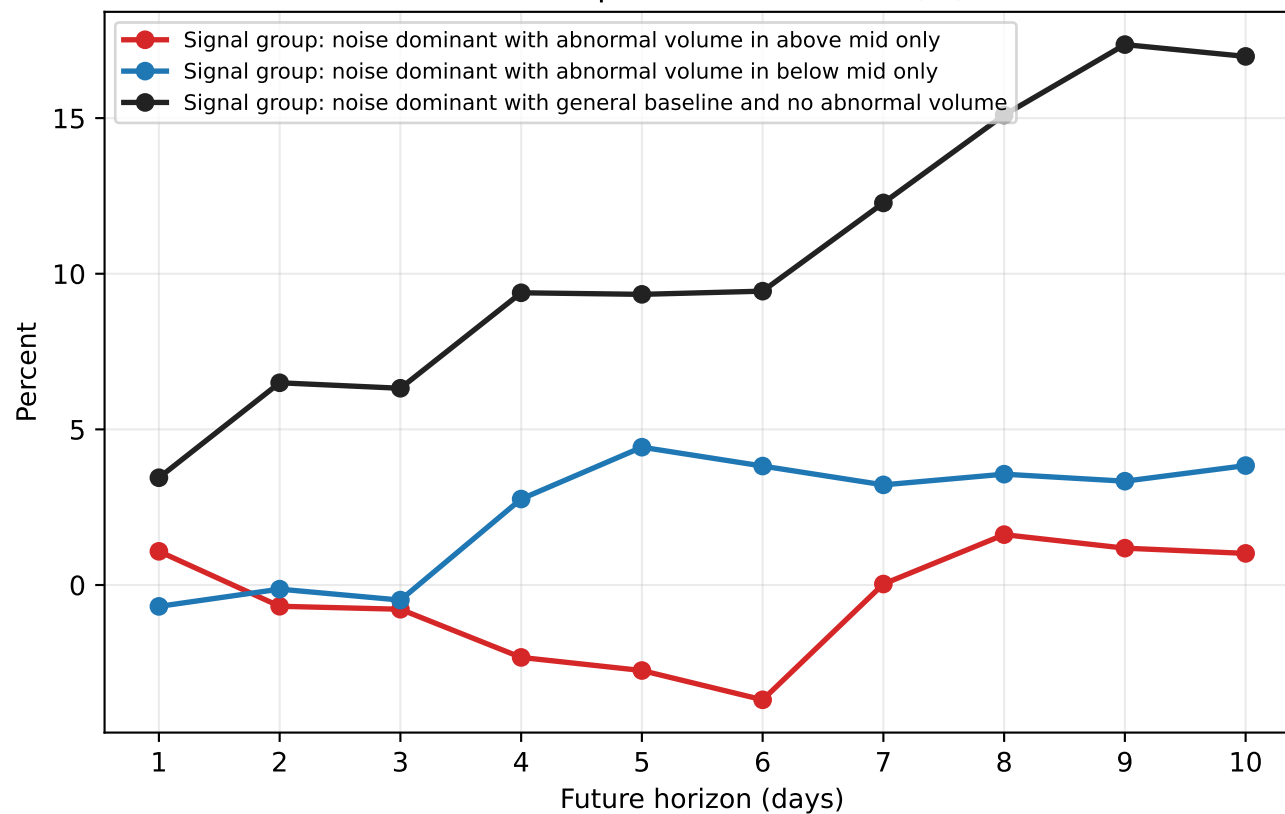
Signal group: detectable downward drift with abnormal volume in above mid only

n=1 | pre_mu=-0.003 | pre_sigma_seq=12.08%

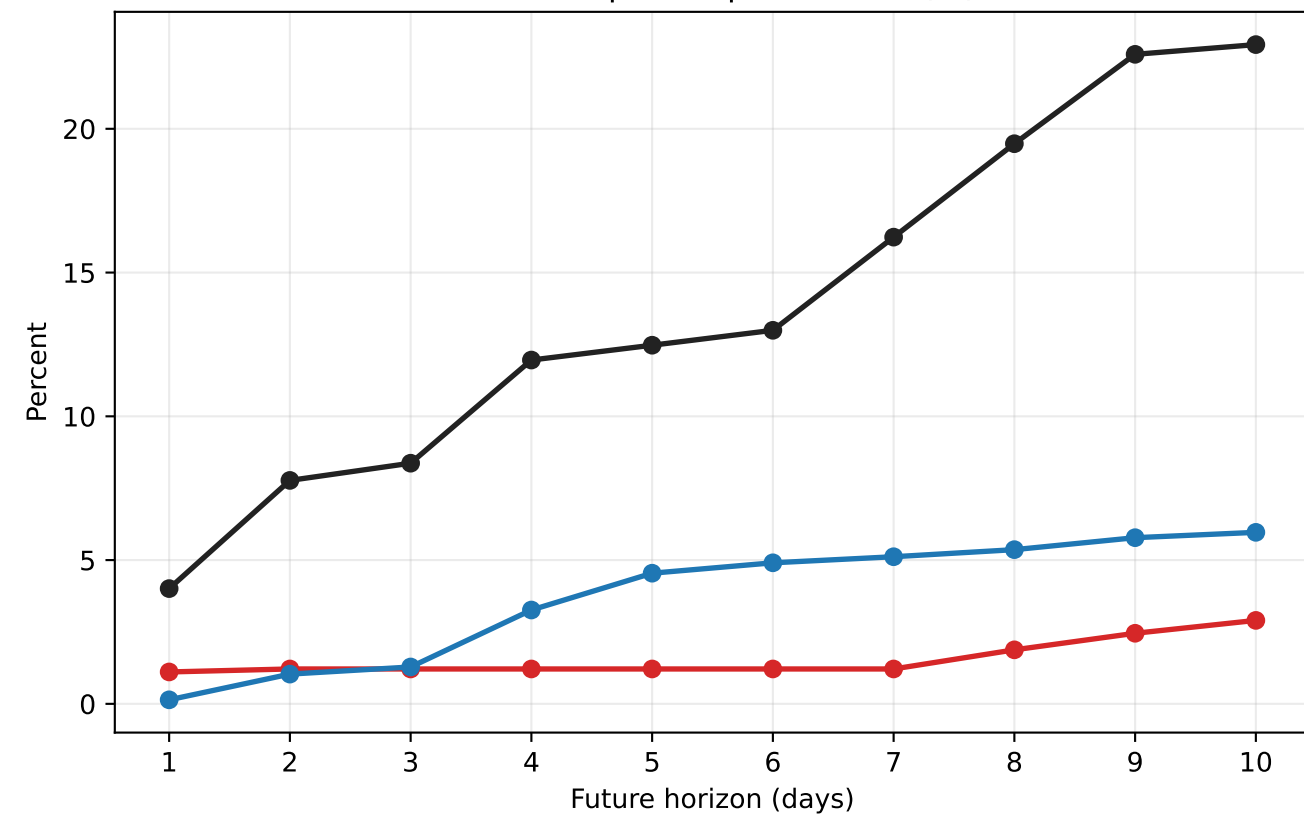
future_mean_return=-3.53% | future_sigma_seq=9.90%

p(mu)=nan | p(sigma)=nan | p(future sigma)=nan

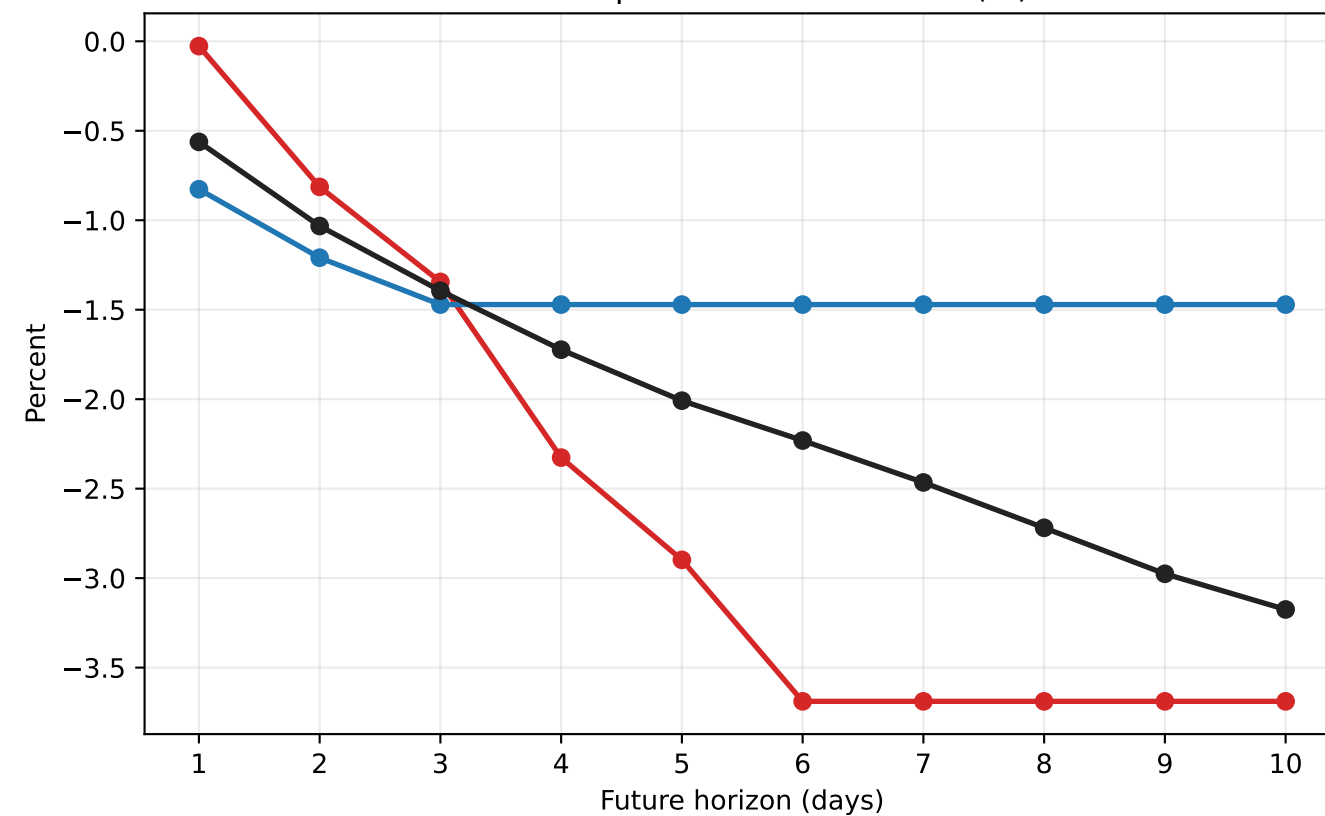
Future seq cumulative return (%)



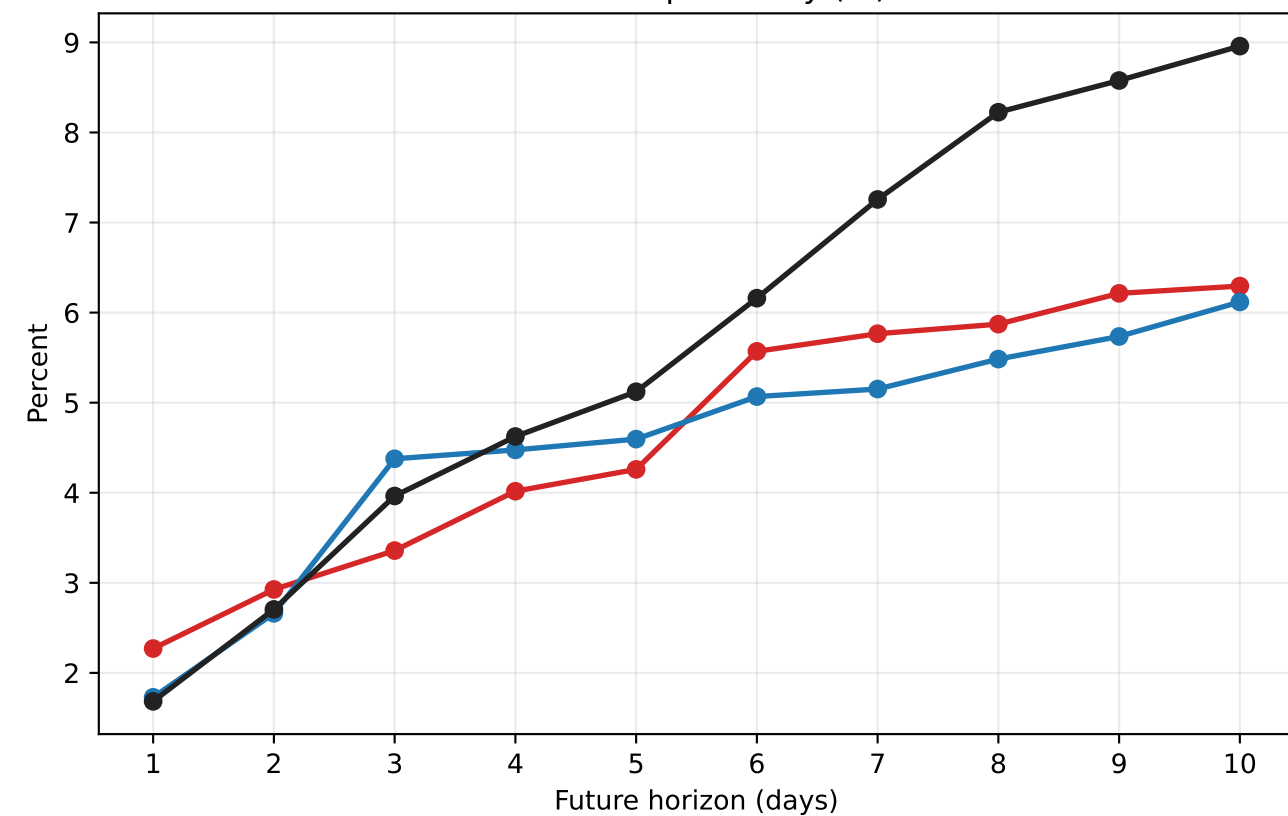
Future seq max up excursion (%)



Future seq max down excursion (%)



Future seq volatility (%)



META Daily Noise Dominant Notes

Daily comparison summary

Reference group = largest-n group at terminal horizon

Reference: Signal group: noise dominant with general baseline and no abnormal volume

Signal group: noise dominant with general baseline and no abnormal volume

n=470 | pre_mu=0.001748 | pre_sigma_seq=7.70%

future_mean_return=16.98% | future_sigma_seq=8.96%

p(mu)=nan | p(sigma)=nan | p(future sigma)=nan

Signal group: noise dominant with abnormal volume in below mid only

n=3 | pre_mu=0.0415 | pre_sigma_seq=66.15%

future_mean_return=3.84% | future_sigma_seq=6.12%

p(mu)=0.401 | p(sigma)=0.447 | p(future sigma)=0.208

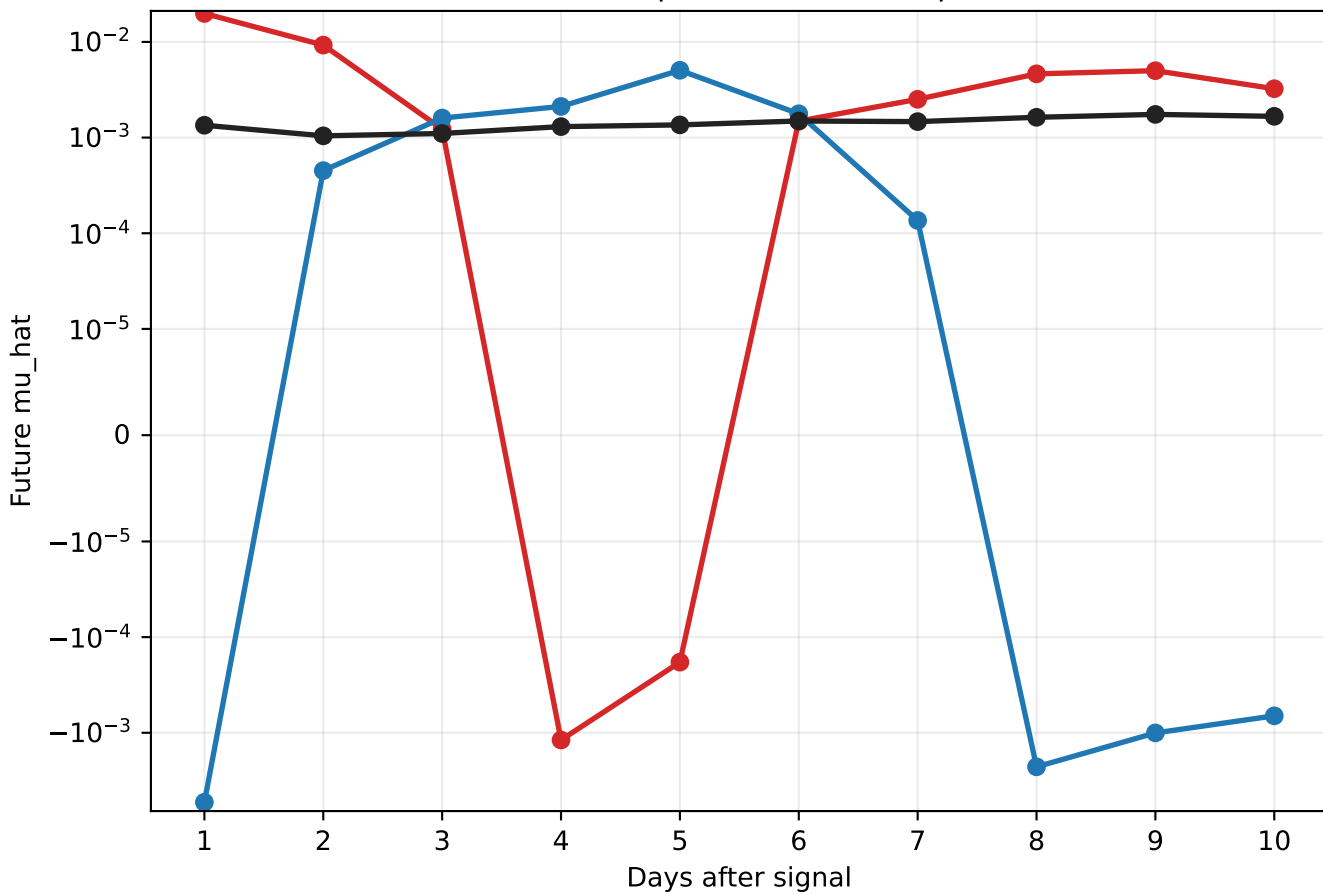
Signal group: noise dominant with abnormal volume in above mid only

n=2 | pre_mu=0.05689 | pre_sigma_seq=97.10%

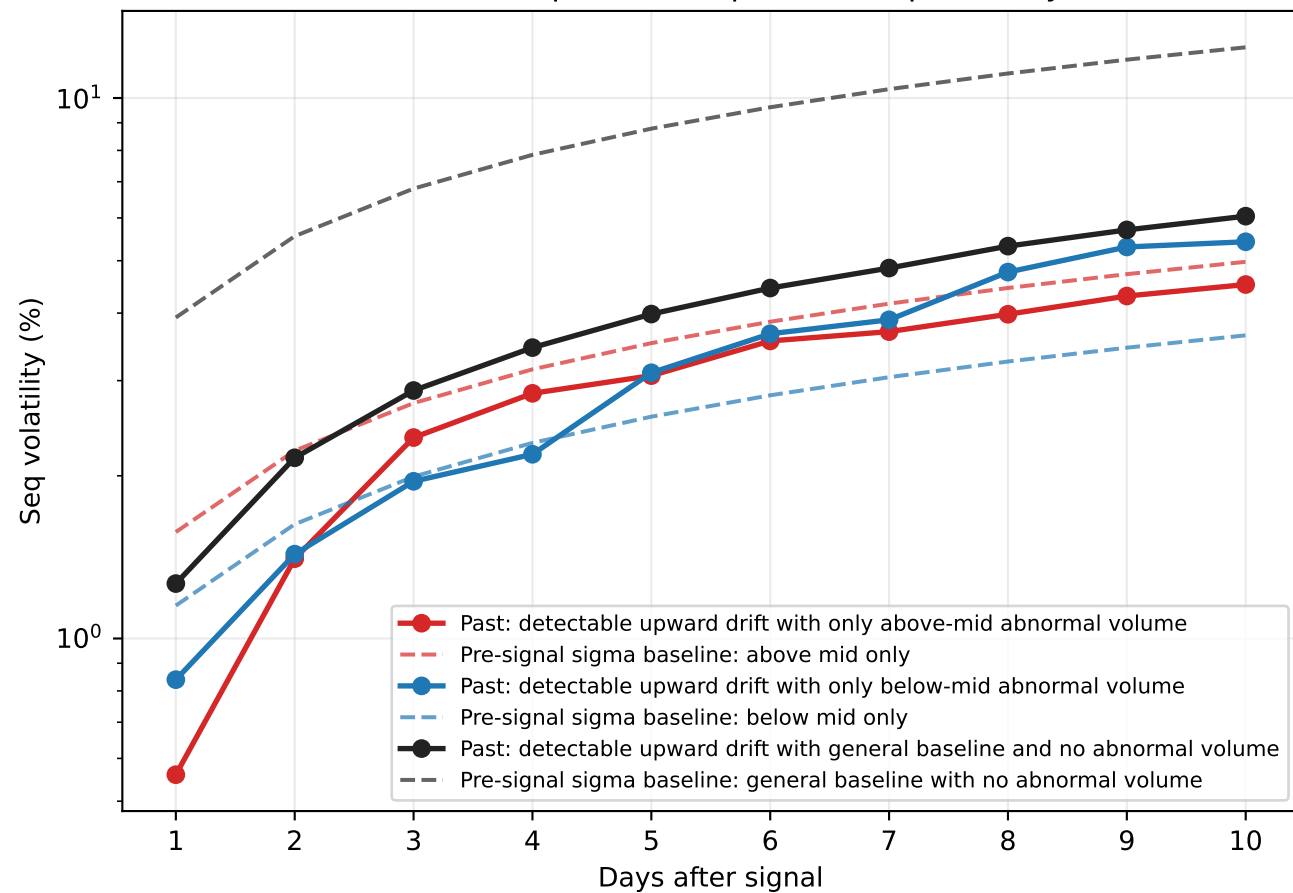
future_mean_return=1.02% | future_sigma_seq=6.29%

p(mu)=0.513 | p(sigma)=0.515 | p(future sigma)=0.501

Future drift parameter mean path



Future diffusion parameter path as seq volatility (%)



META Daily Detectable Upward Drift Parameter Notes

Daily detectable upward drift parameter comparison

Groups: above-mid only vs below-mid only vs no abnormal baseline

Past: detectable upward drift with only above-mid abnormal volume

n=1 | pre_mu=0.002321 | pre_sigma_seq=4.98%

post_mu(10)=0.003243 | post_sigma_seq(10)=4.52%

p_mu_vs_baseline=nan | p_sigma_vs_baseline=nan

Past: detectable upward drift with only below-mid abnormal volume

n=3 | pre_mu=0.003783 | pre_sigma_seq=3.64%

post_mu(10)=-0.0006649 | post_sigma_seq(10)=5.42%

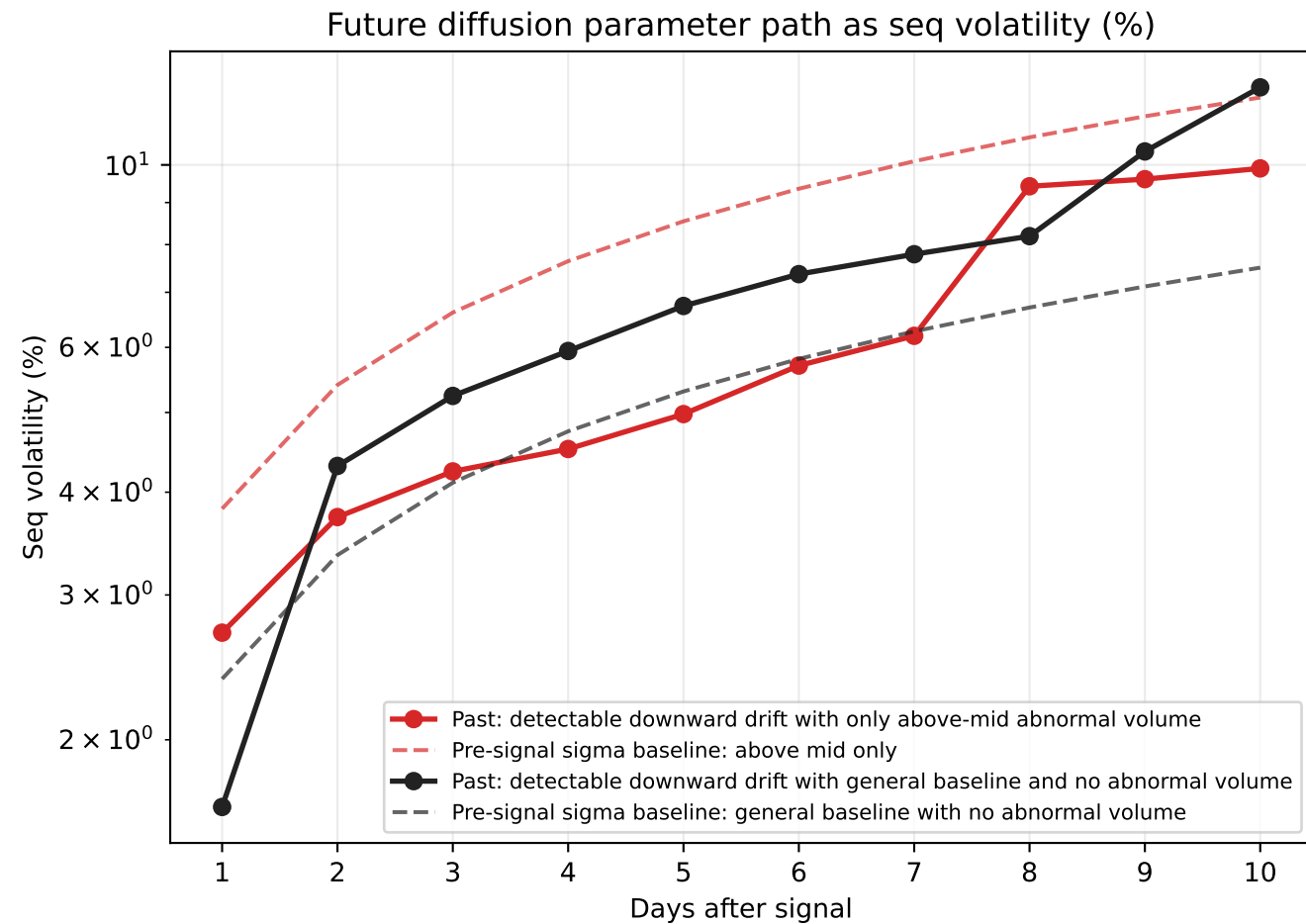
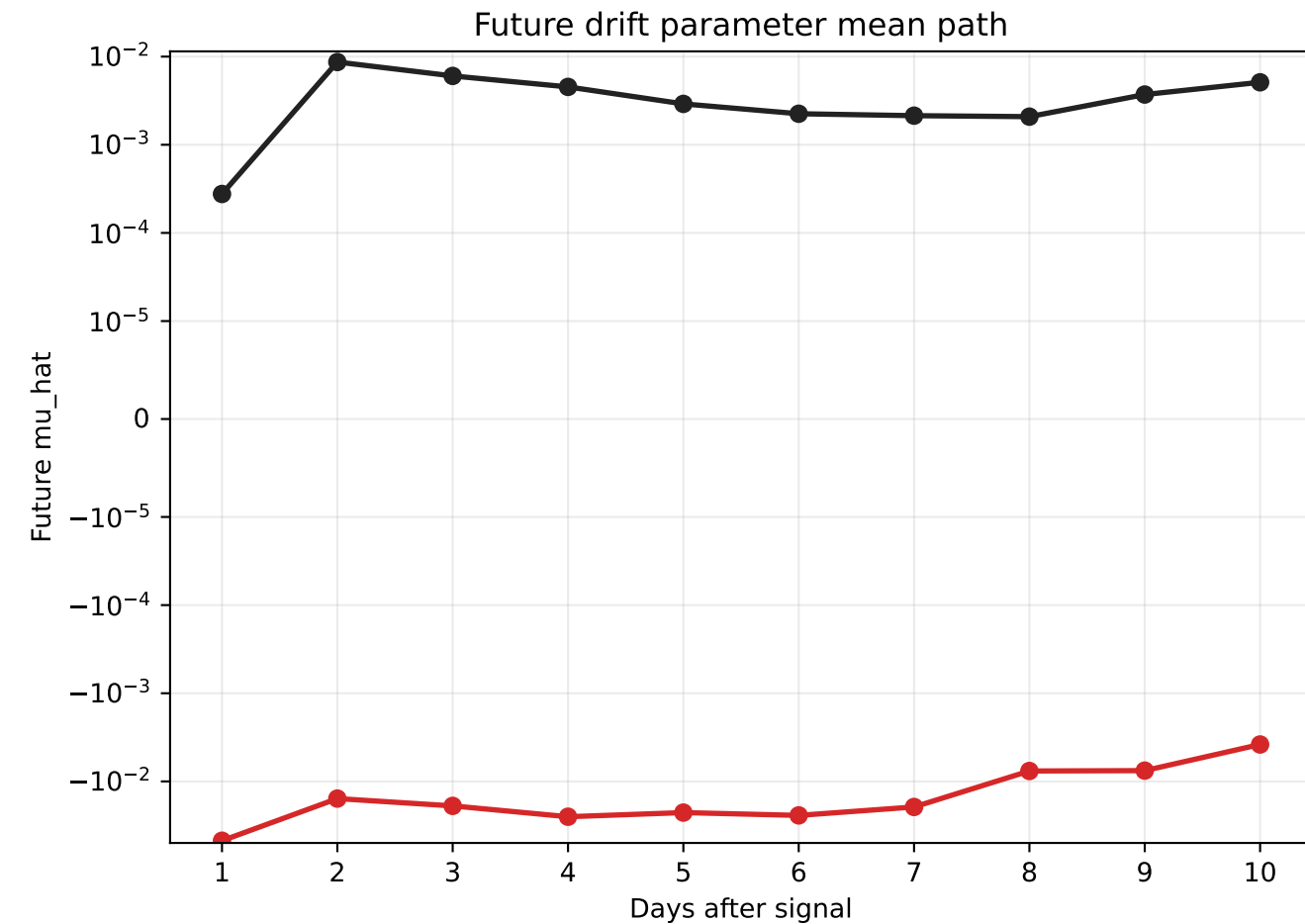
p_mu_vs_baseline=0.0225 | p_sigma_vs_baseline=0.292

Past: detectable upward drift with general baseline and no abnormal volume

n=311 | pre_mu=0.008237 | pre_sigma_seq=12.41%

post_mu(10)=0.001668 | post_sigma_seq(10)=6.05%

p_mu_vs_baseline=nan | p_sigma_vs_baseline=nan



META Daily Detectable Downward Drift Parameter Notes

Daily detectable downward drift parameter comparison

Groups: above-mid only vs below-mid only vs no abnormal baseline

Past: detectable downward drift with only above-mid abnormal volume

n=1 | pre_mu=-0.003 | pre_sigma_seq=12.08%

post_mu(10)=-0.003812 | post_sigma_seq(10)=9.90%

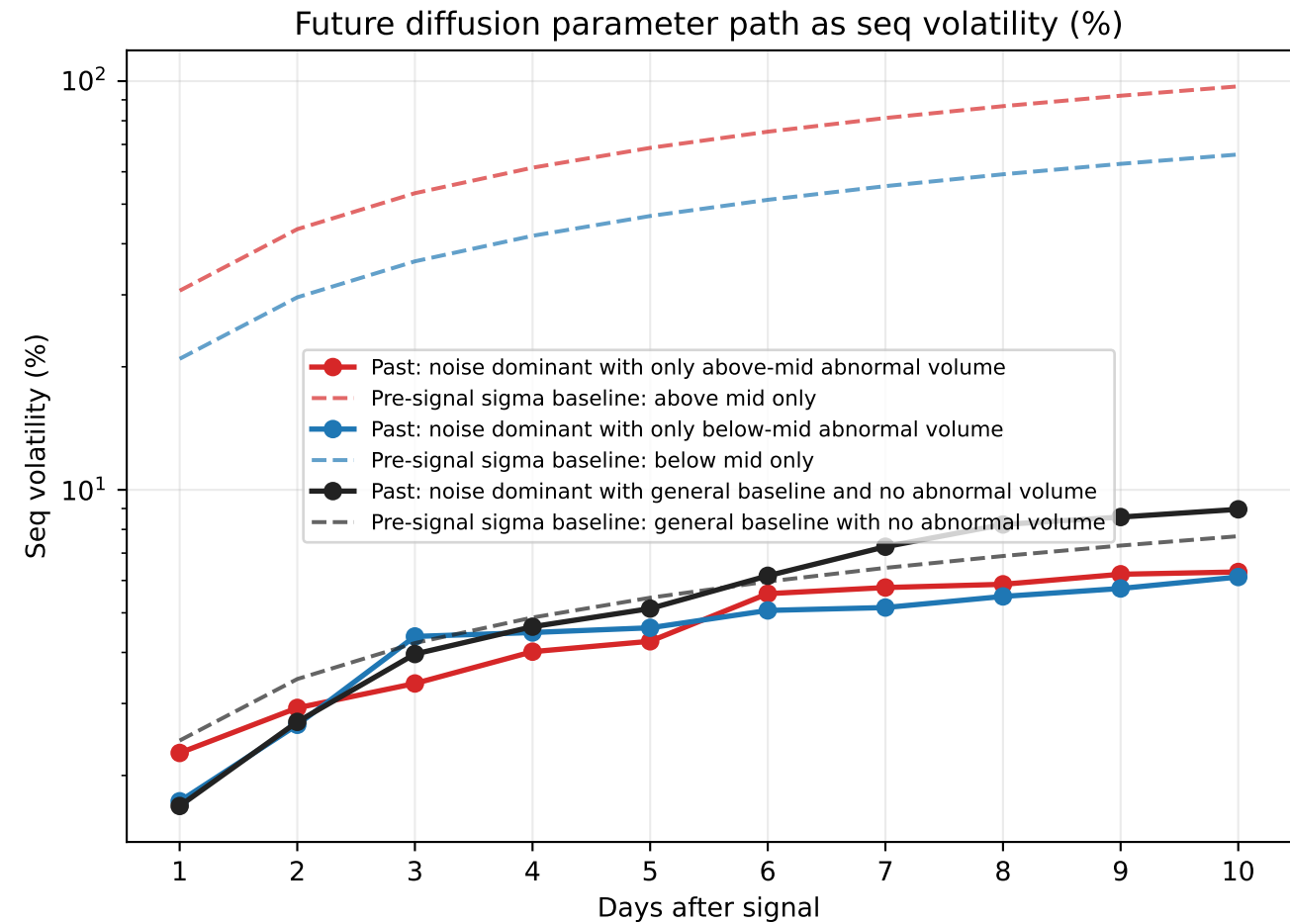
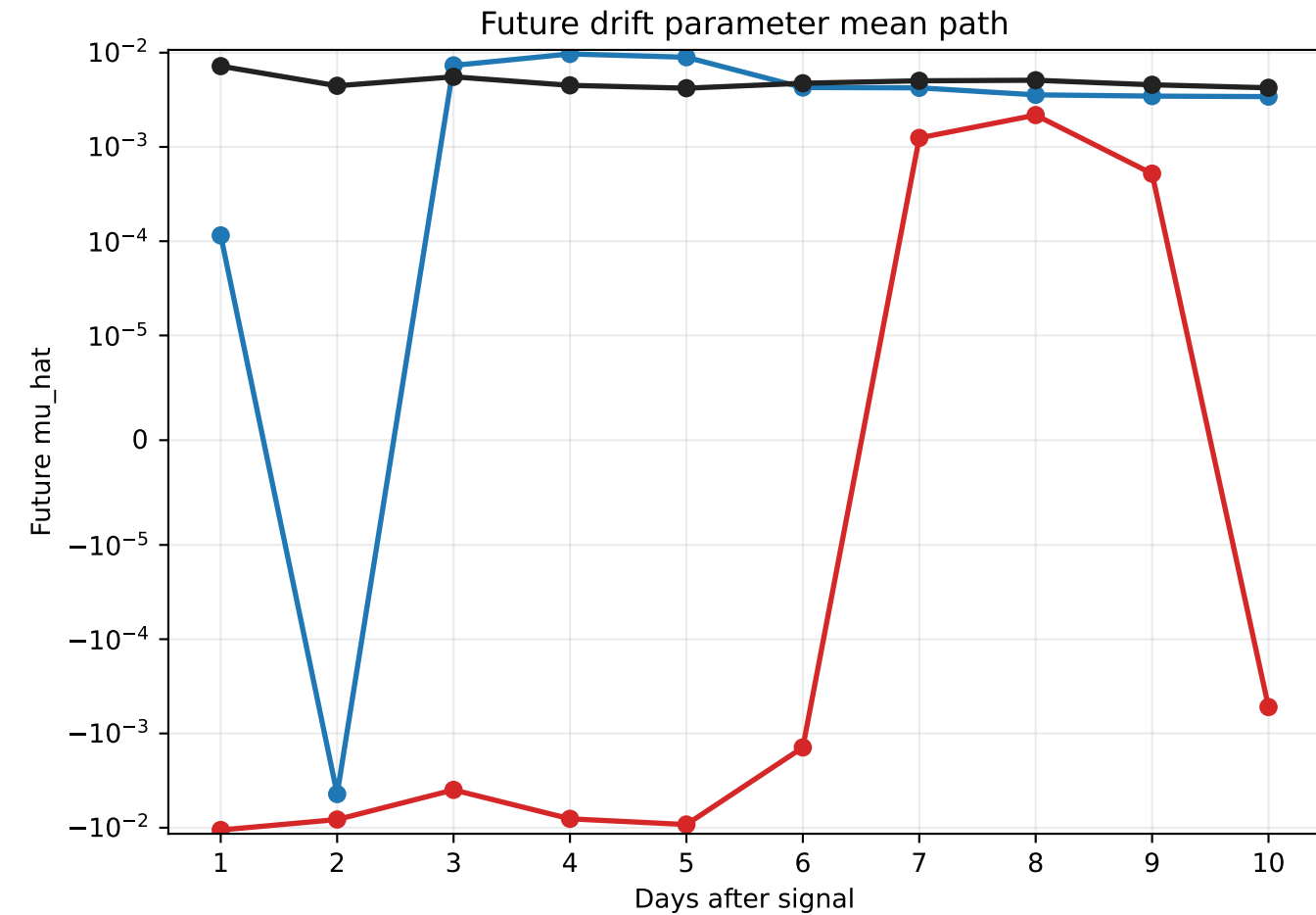
p_mu_vs_baseline=nan | p_sigma_vs_baseline=nan

Past: detectable downward drift with general baseline and no abnormal volume

n=163 | pre_mu=-0.003916 | pre_sigma_seq=7.50%

post_mu(10)=0.005111 | post_sigma_seq(10)=12.42%

p_mu_vs_baseline=nan | p_sigma_vs_baseline=nan



META Daily Noise Dominant Parameter Notes

Daily noise dominant parameter comparison

Groups: above-mid only vs below-mid only vs no abnormal baseline

Past: noise dominant with only above-mid abnormal volume

n=2 | pre_mu=0.05689 | pre_sigma_seq=97.10%

post_mu(10)=-0.0005246 | post_sigma_seq(10)=6.29%

p_mu_vs_baseline=0.207 | p_sigma_vs_baseline=0.501

Past: noise dominant with only below-mid abnormal volume

n=3 | pre_mu=0.0415 | pre_sigma_seq=66.15%

post_mu(10)=0.003418 | post_sigma_seq(10)=6.12%

p_mu_vs_baseline=0.696 | p_sigma_vs_baseline=0.208

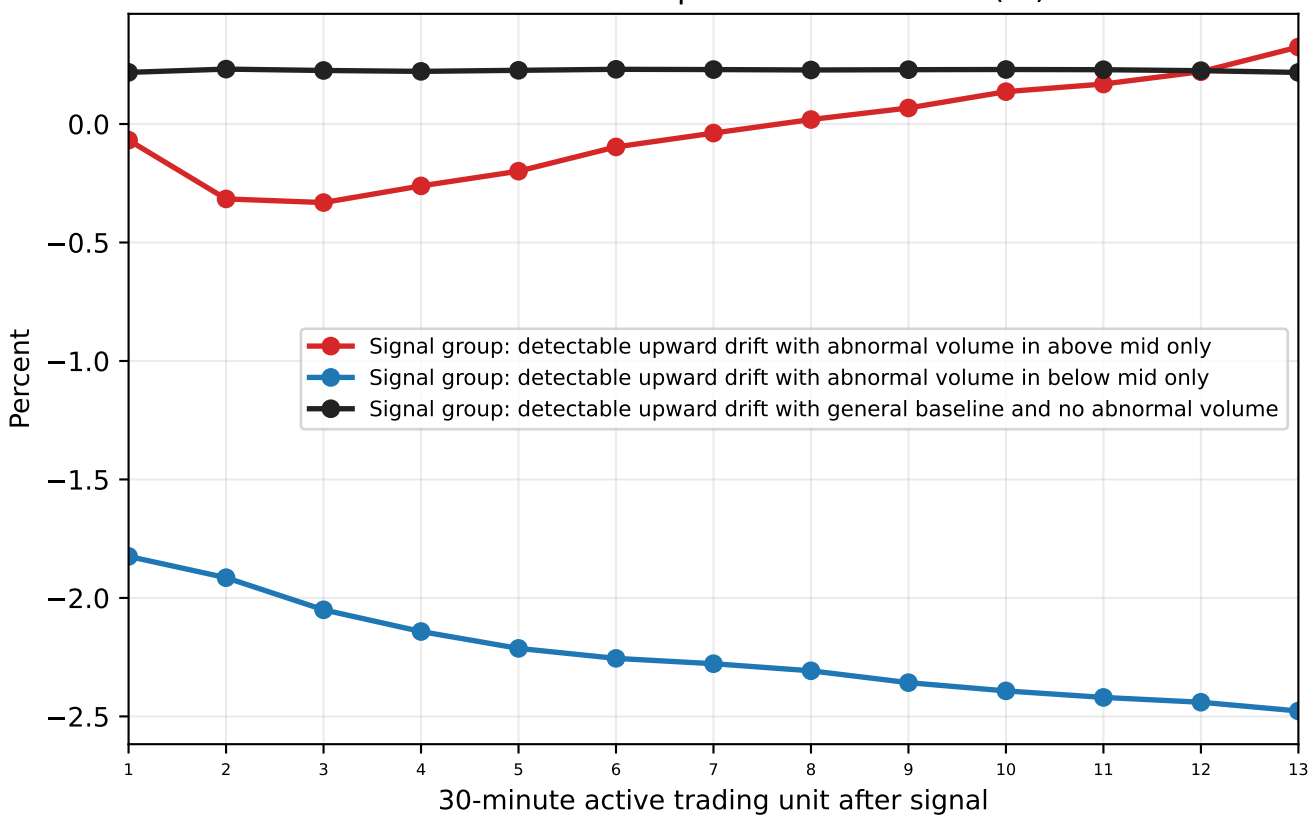
Past: noise dominant with general baseline and no abnormal volume

n=470 | pre_mu=0.001748 | pre_sigma_seq=7.70%

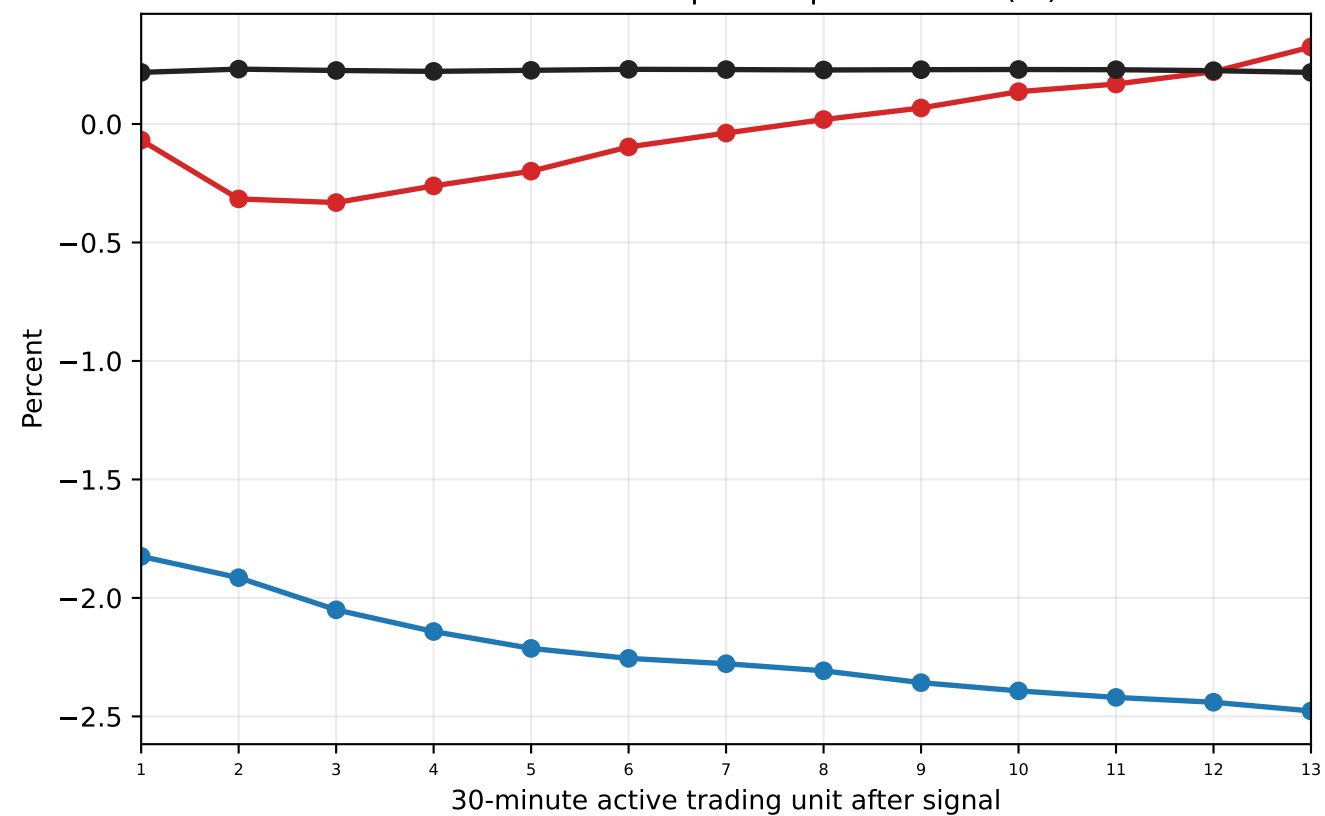
post_mu(10)=0.004239 | post_sigma_seq(10)=8.96%

p_mu_vs_baseline=nan | p_sigma_vs_baseline=nan

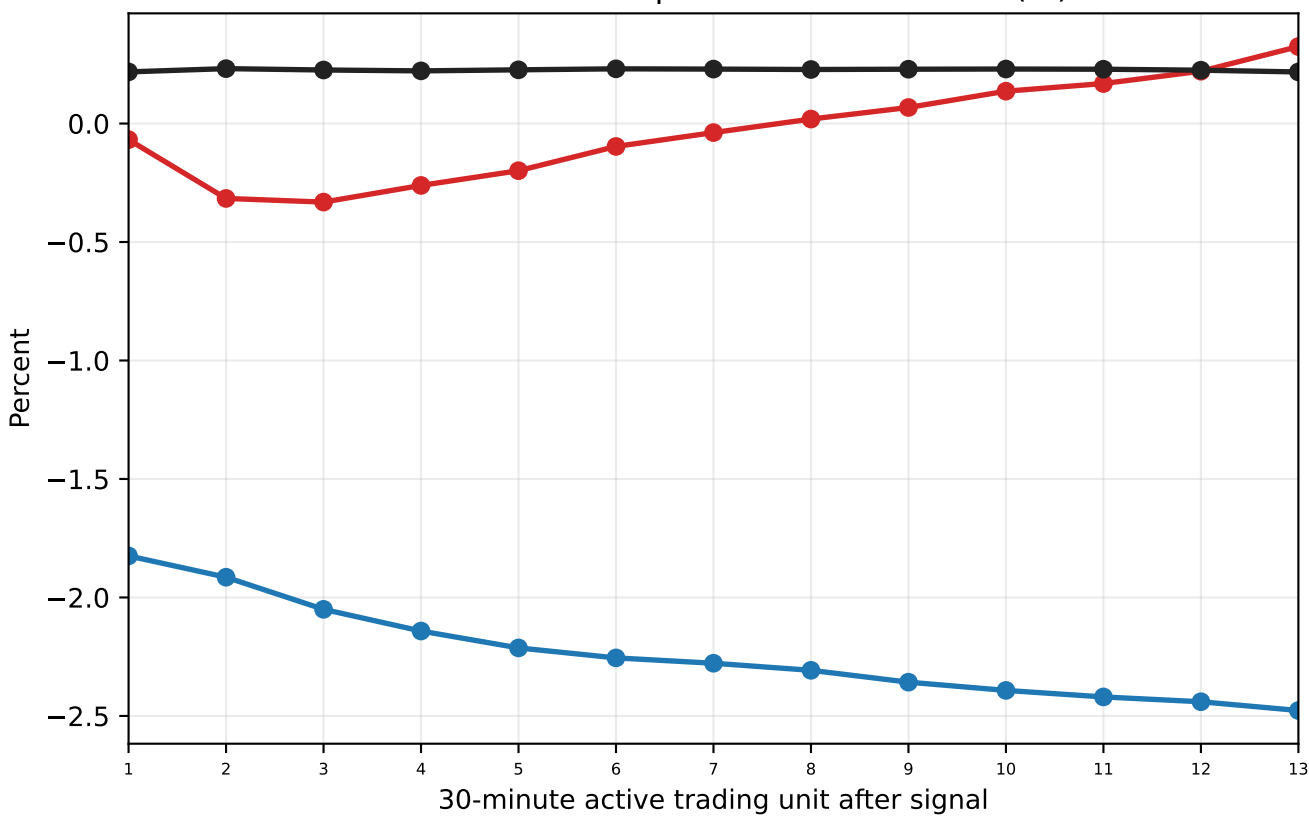
Zoomed future seq cumulative return (%)



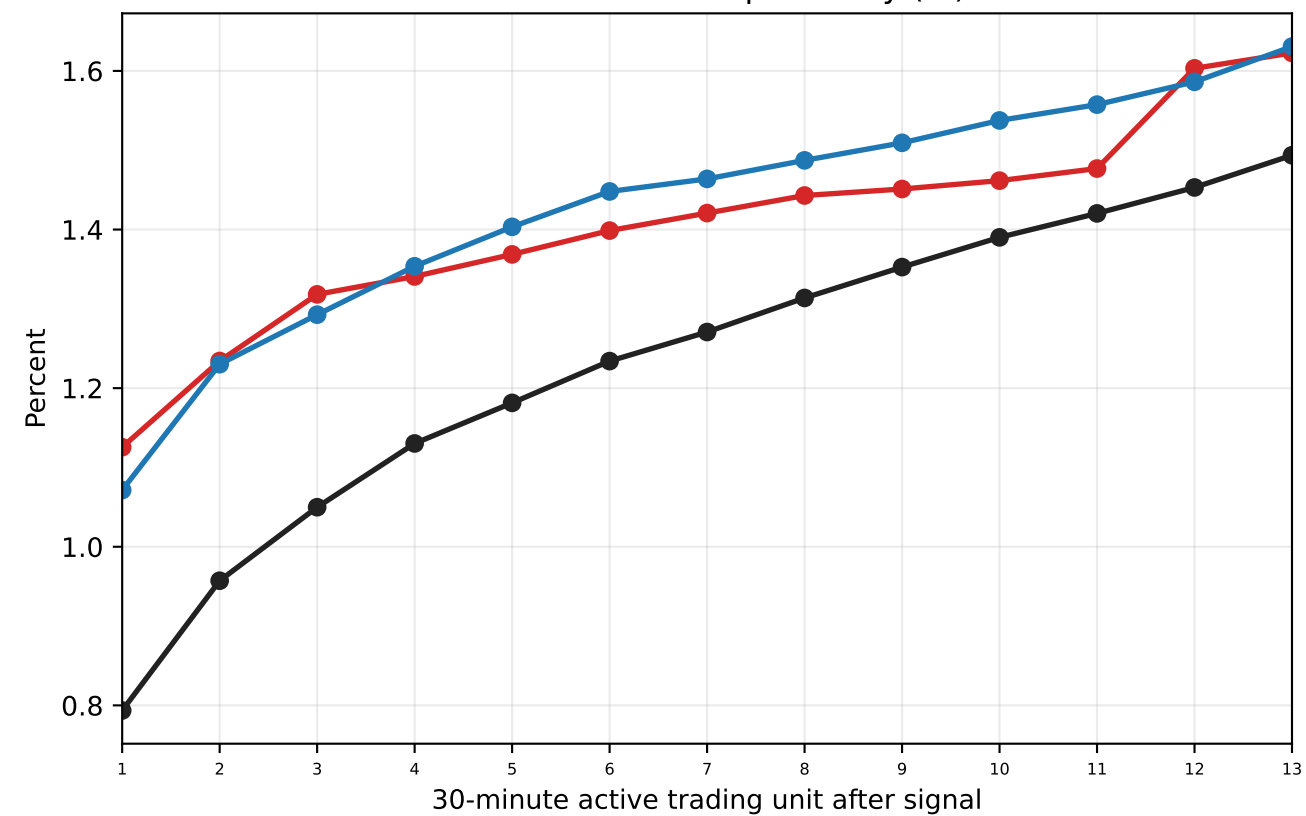
Zoomed future seq max up excursion (%)



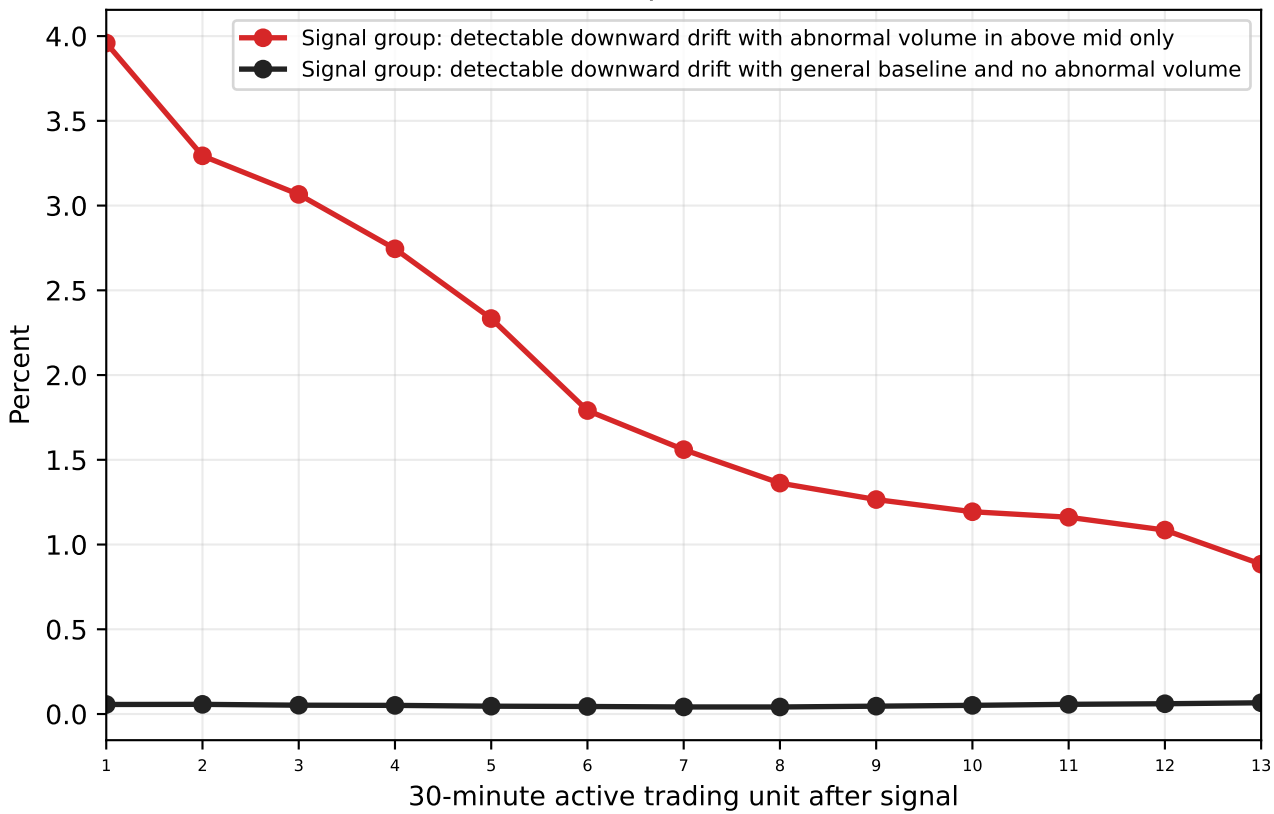
Zoomed future seq max down excursion (%)



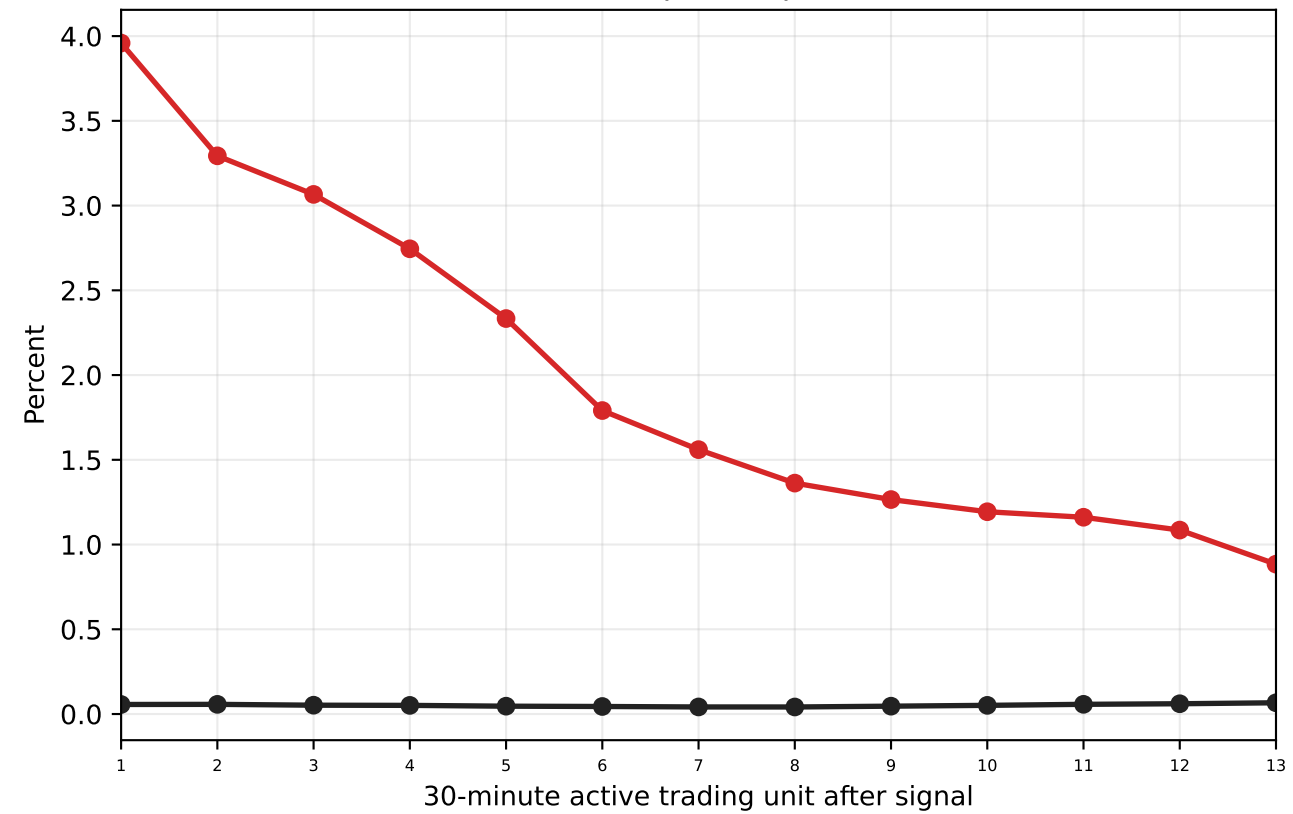
Zoomed future seq volatility (%)



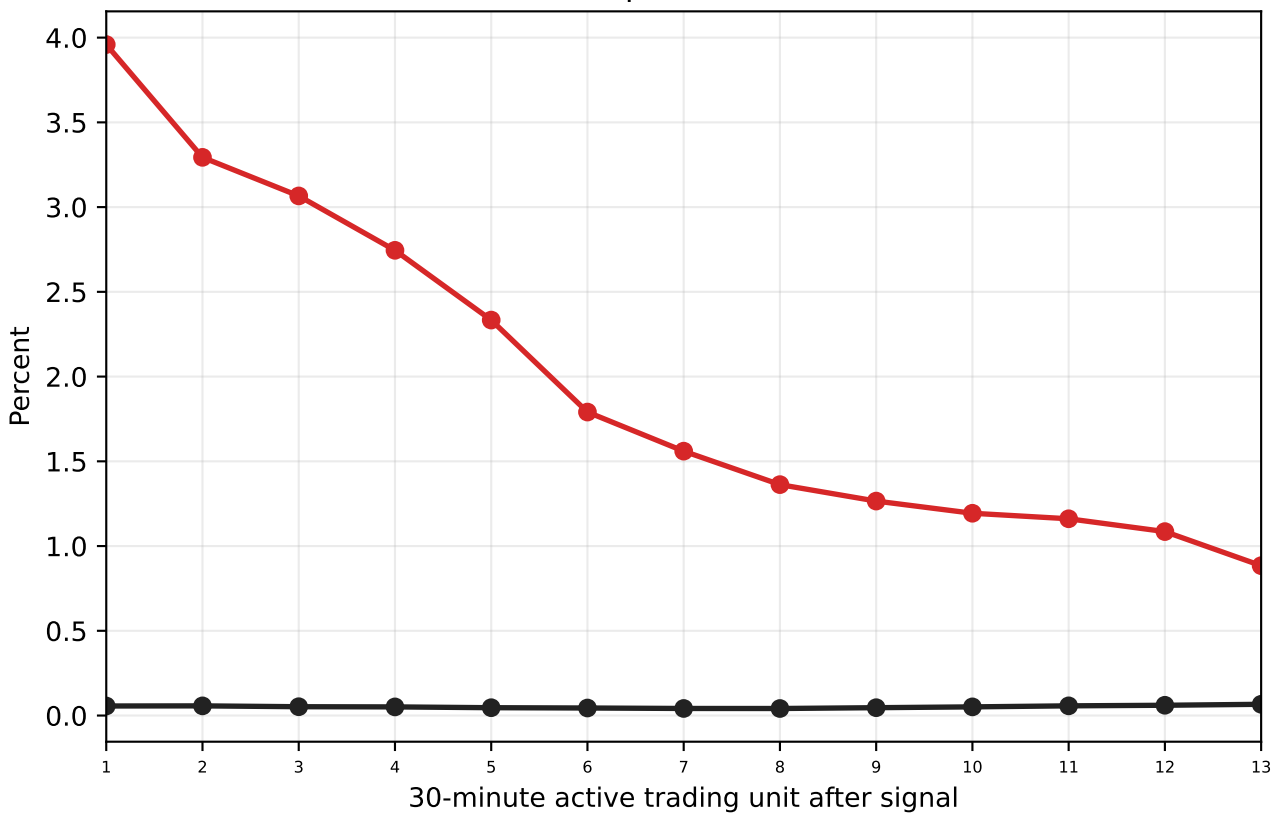
Zoomed future seq cumulative return (%)



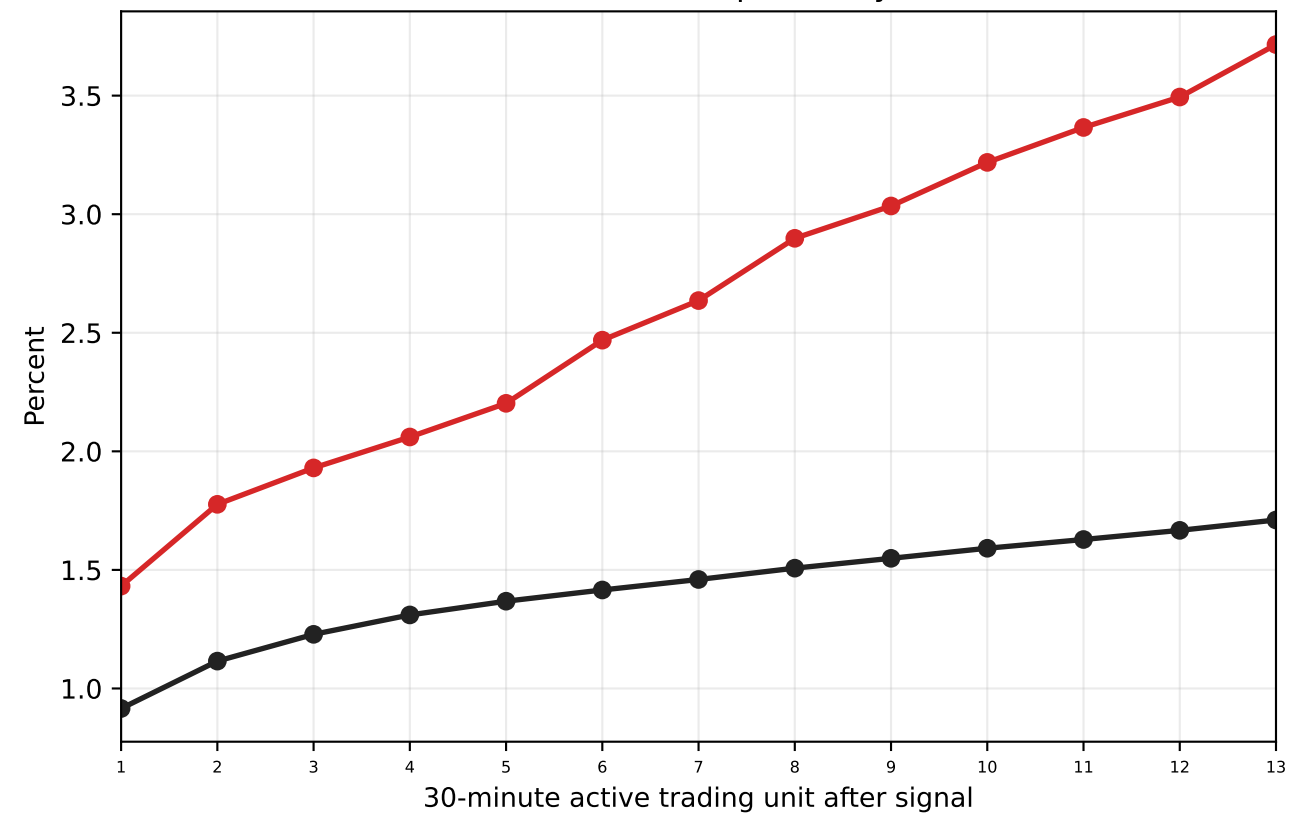
Zoomed future seq max up excursion (%)



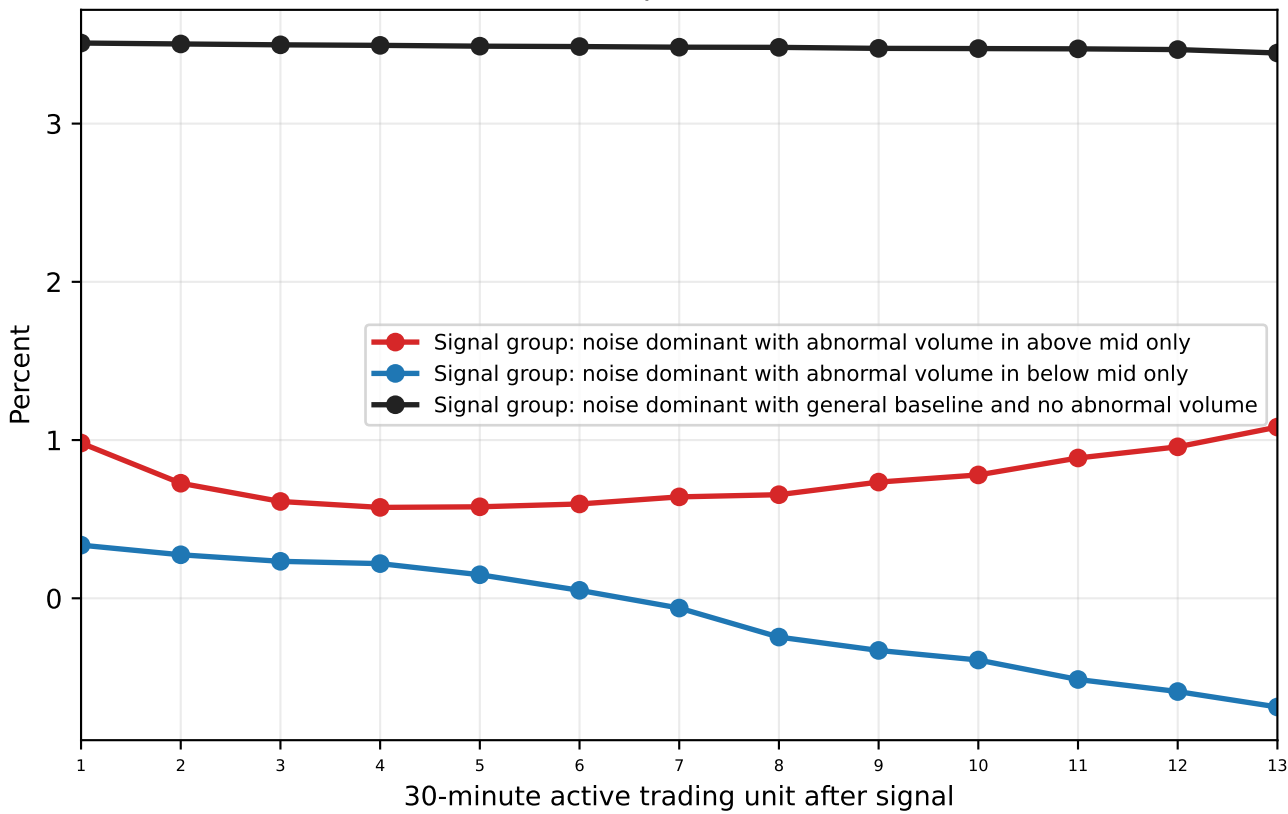
Zoomed future seq max down excursion (%)



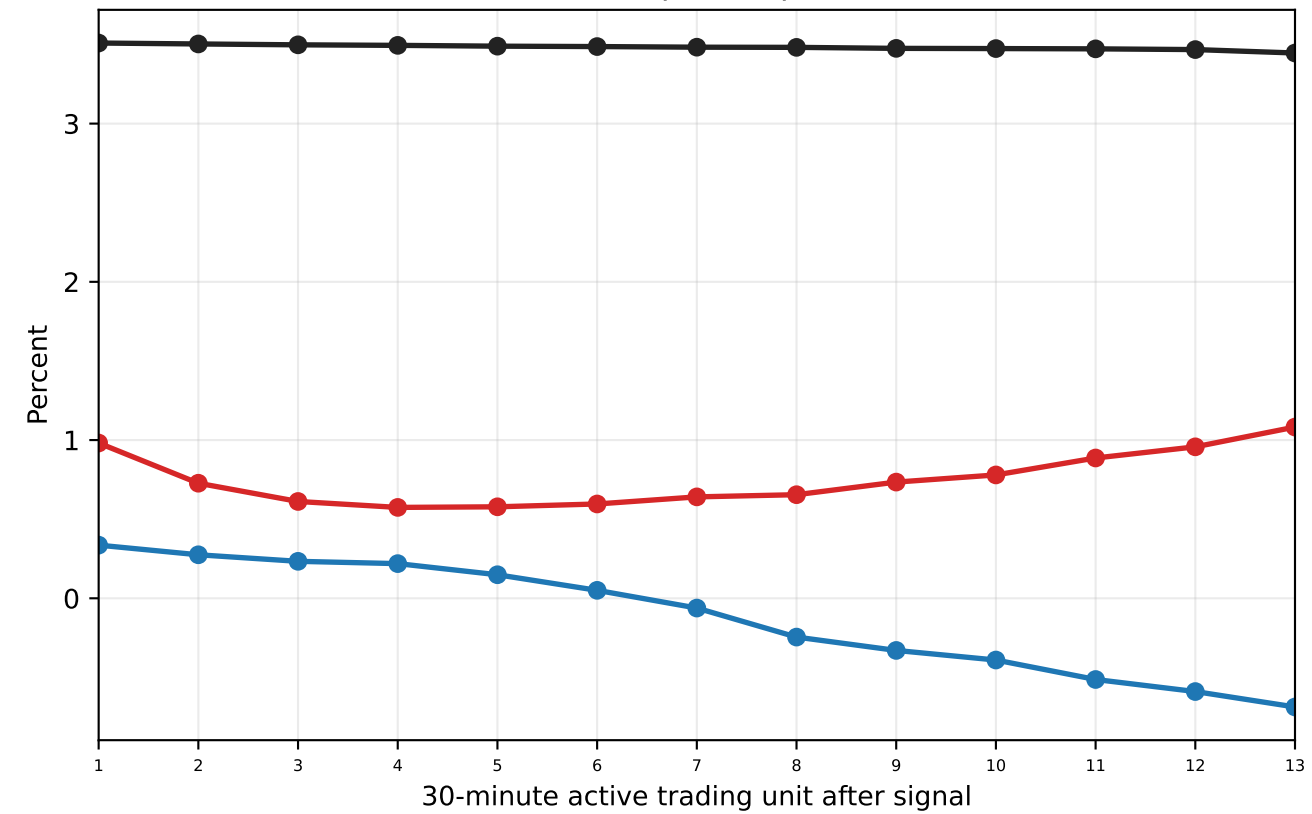
Zoomed future seq volatility (%)



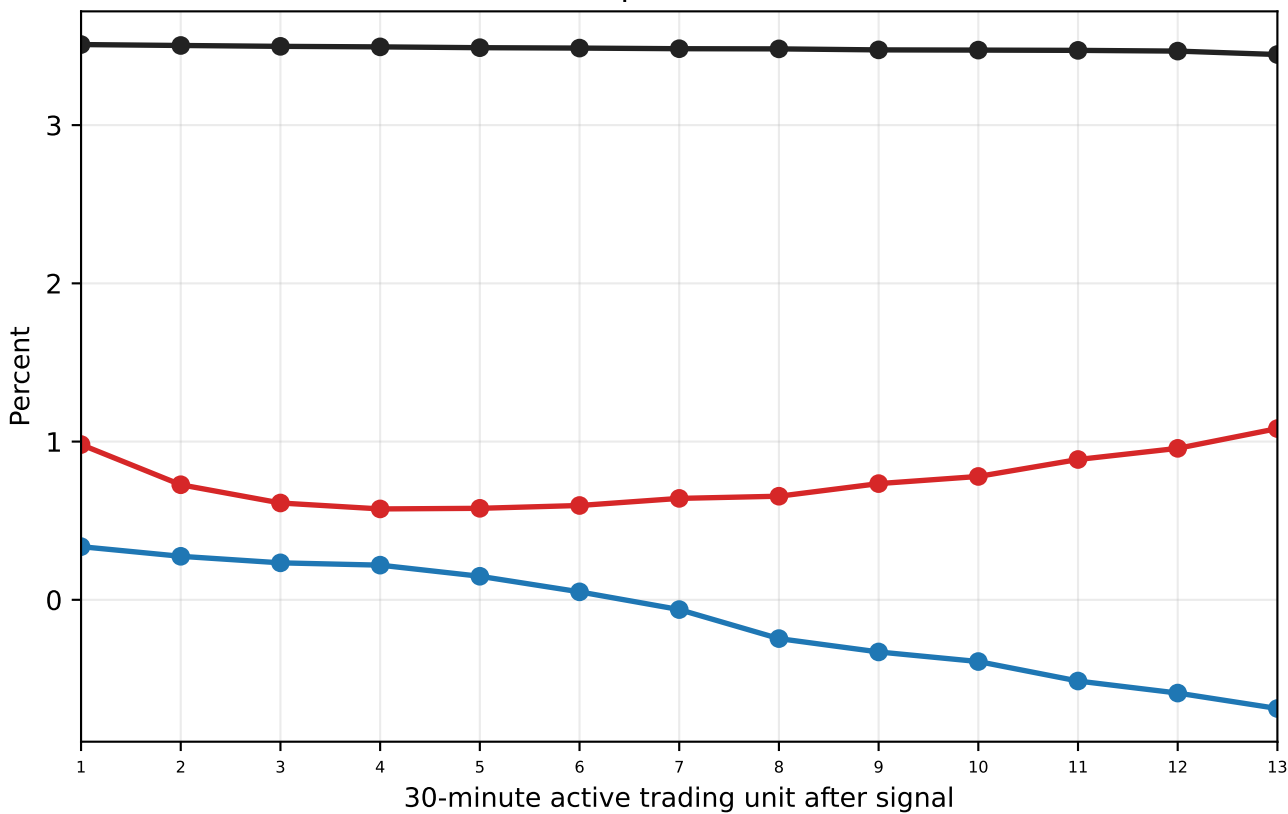
Zoomed future seq cumulative return (%)



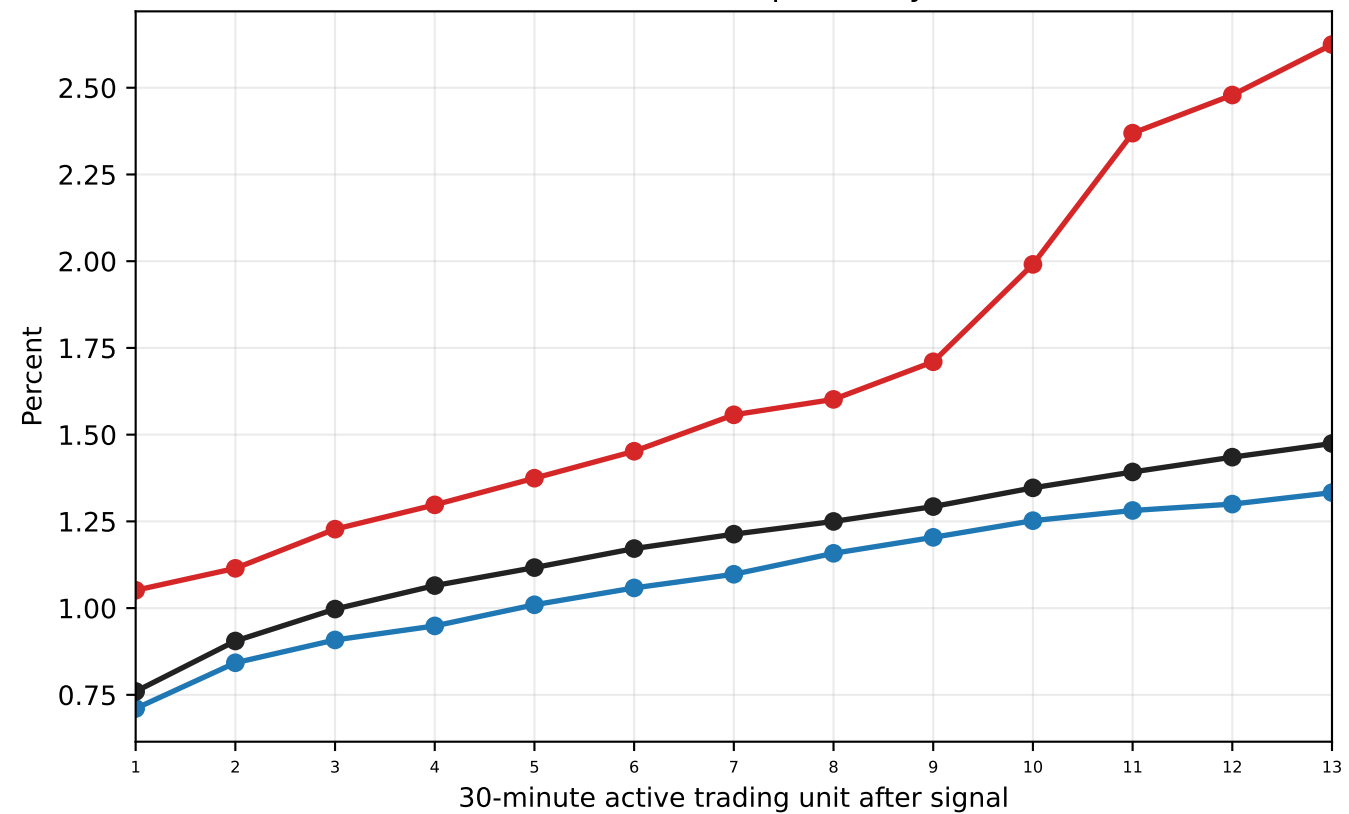
Zoomed future seq max up excursion (%)



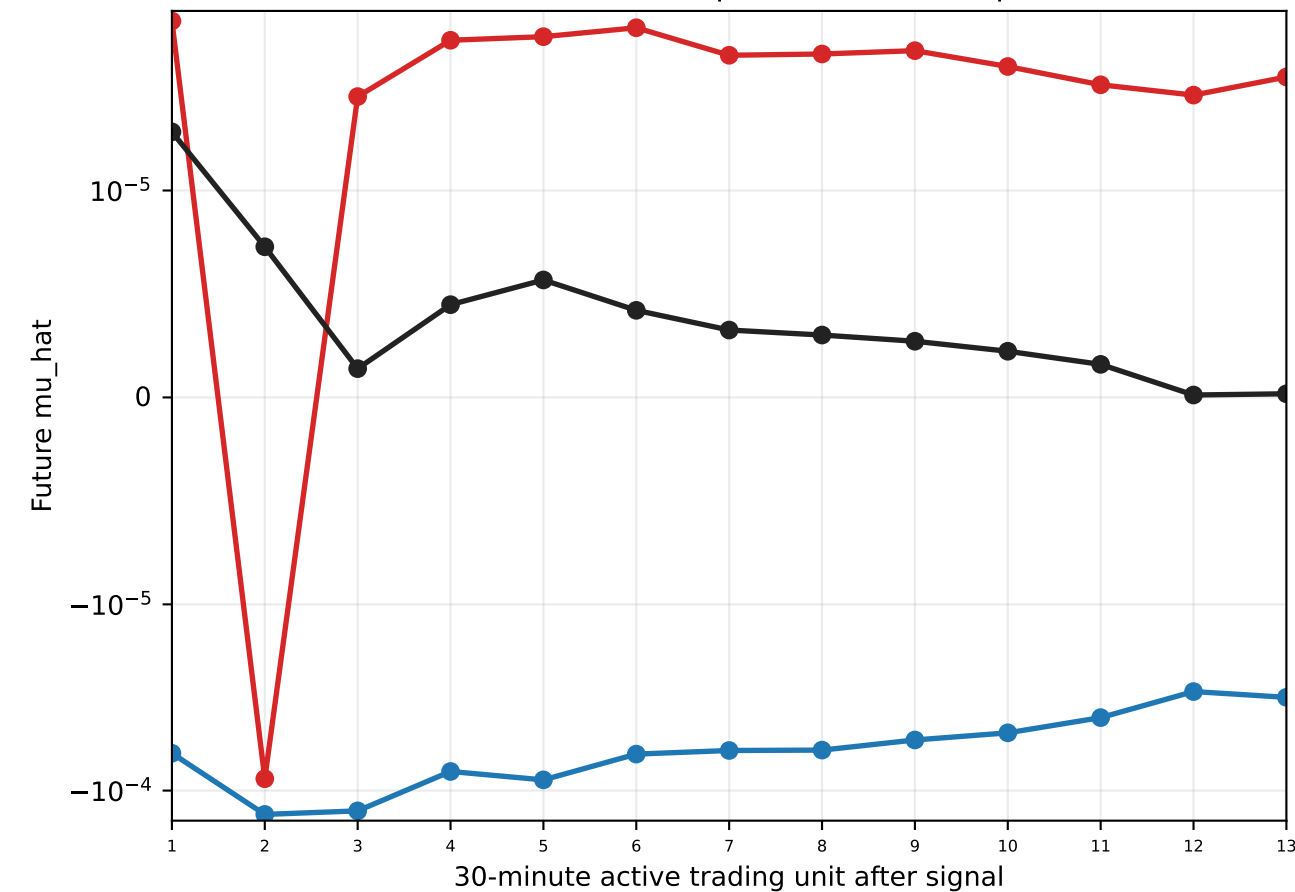
Zoomed future seq max down excursion (%)



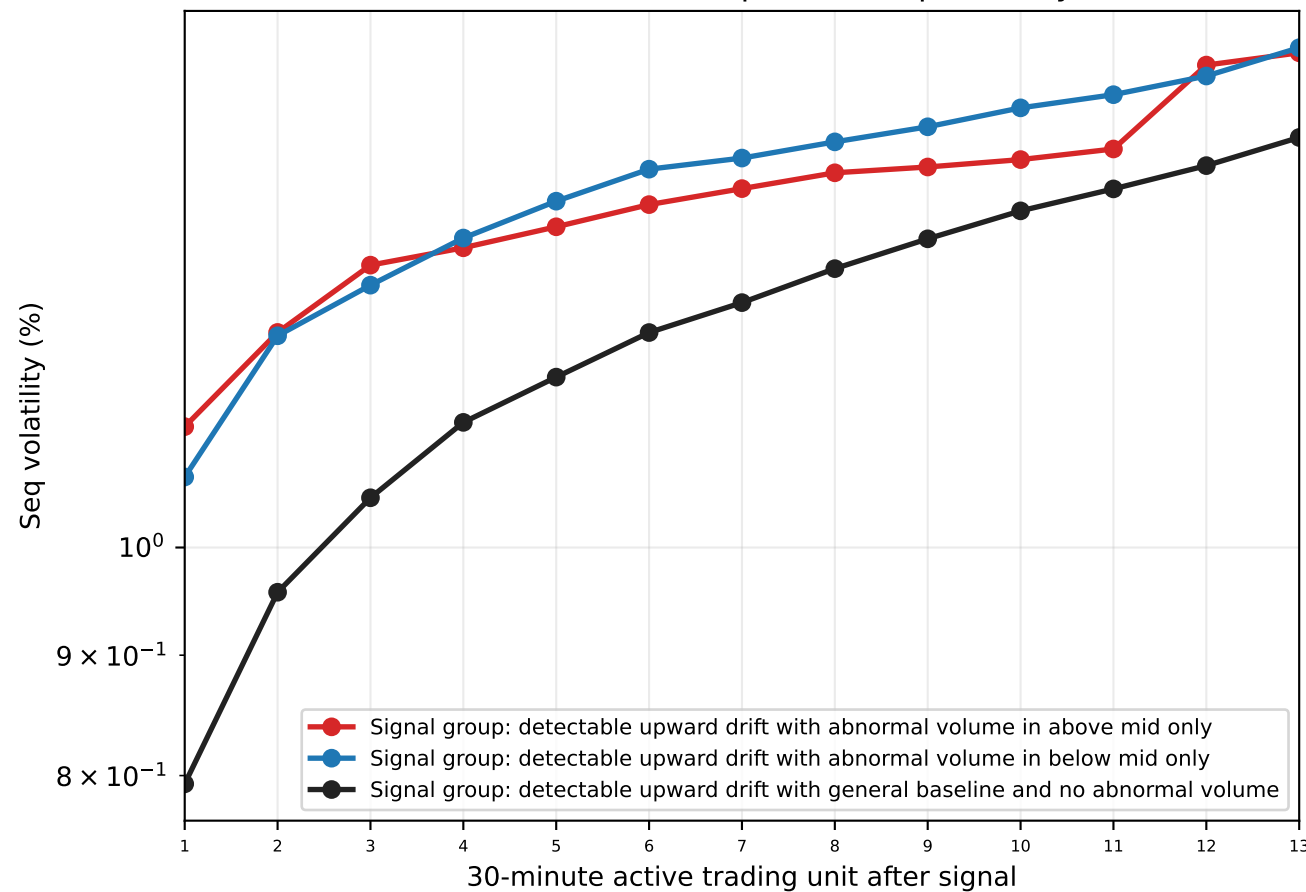
Zoomed future seq volatility (%)



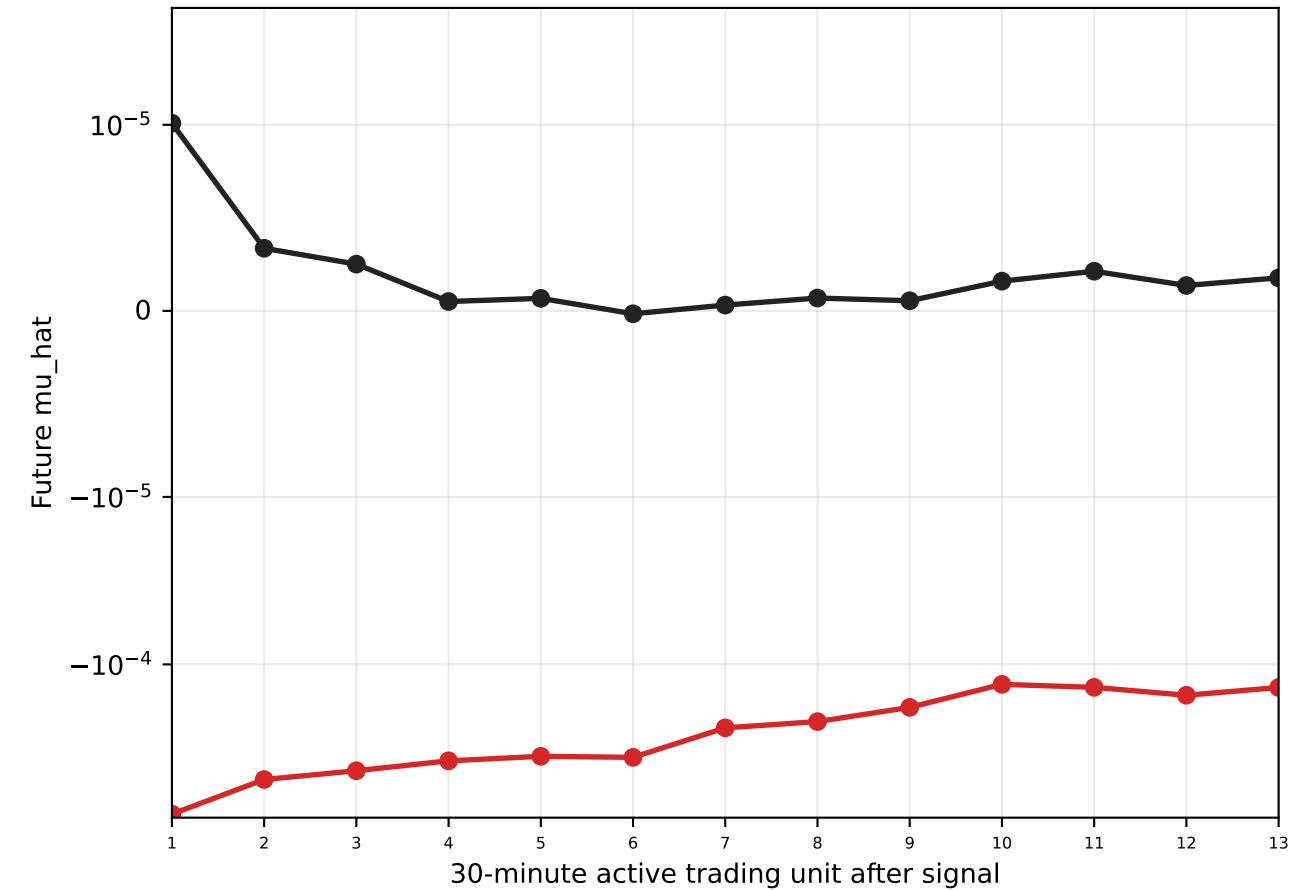
Zoomed future drift parameter mean path



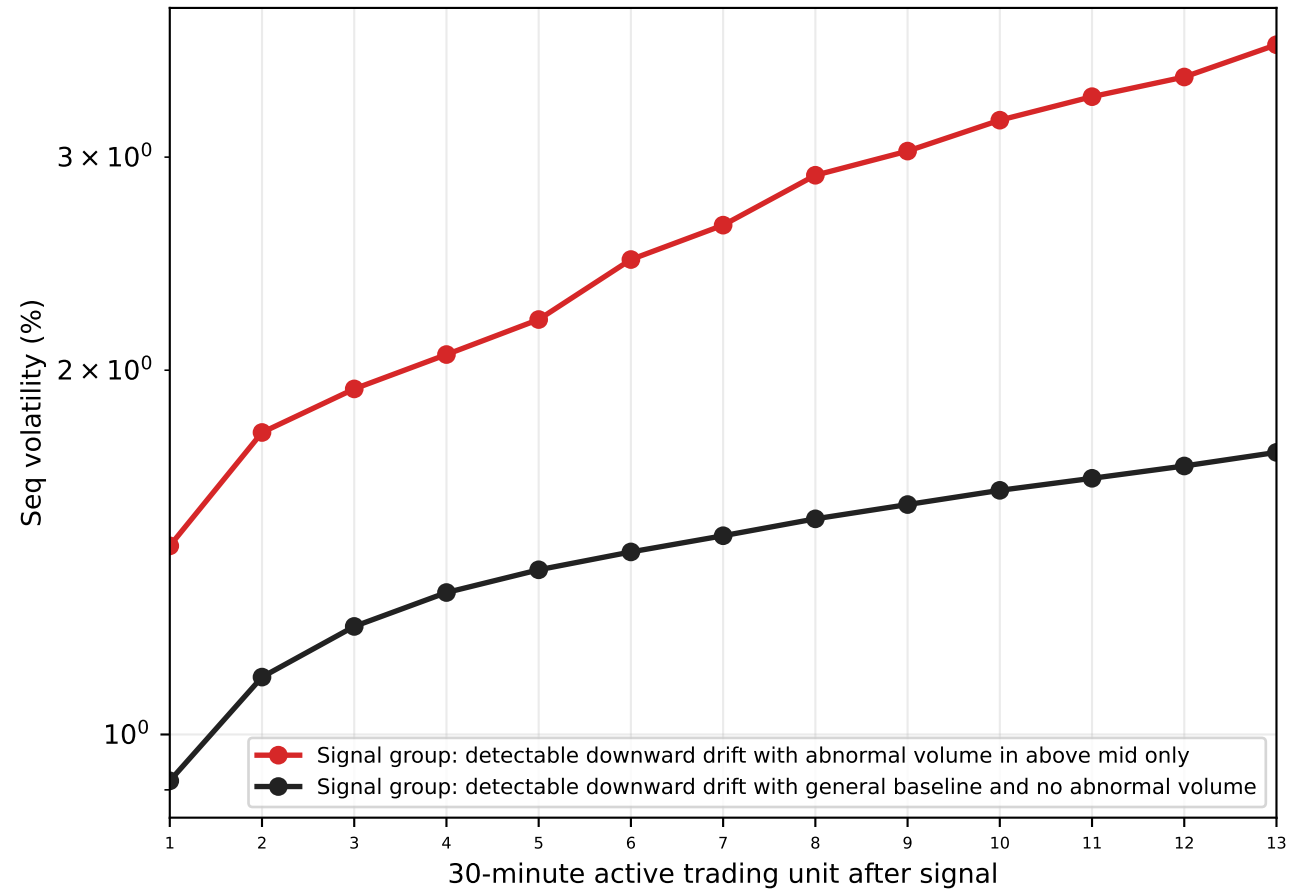
Zoomed future diffusion path as seq volatility (%)



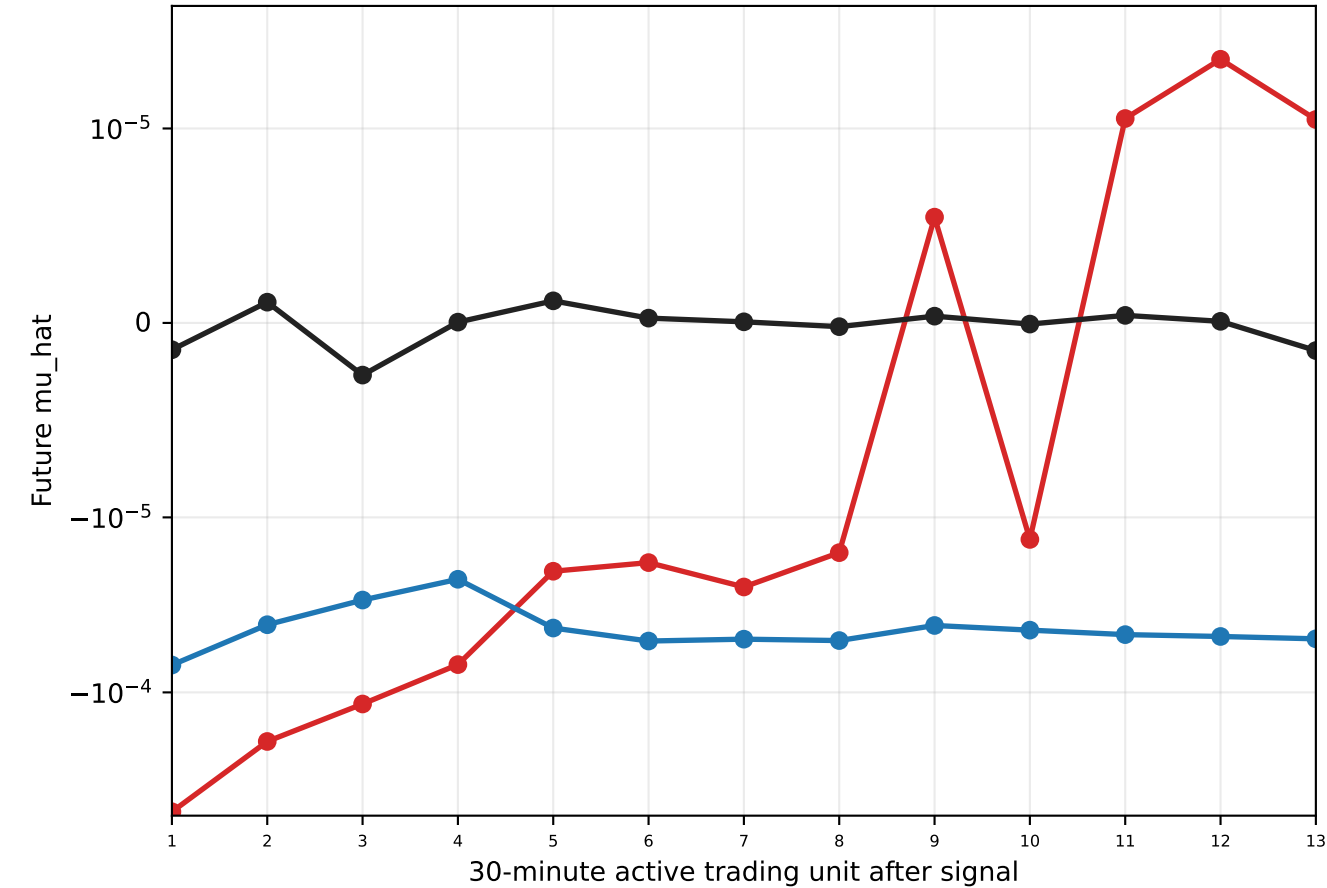
Zoomed future drift parameter mean path



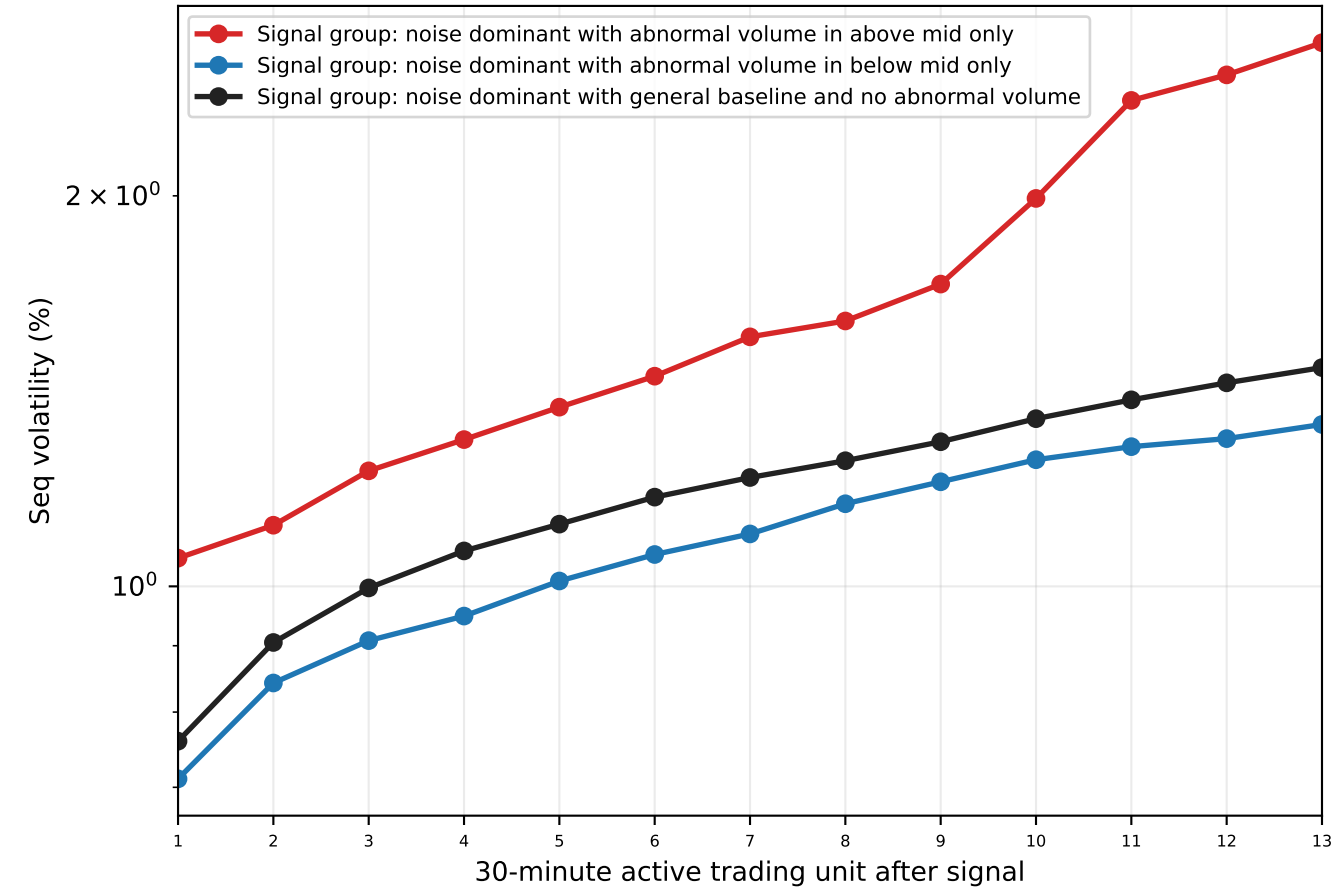
Zoomed future diffusion path as seq volatility (%)



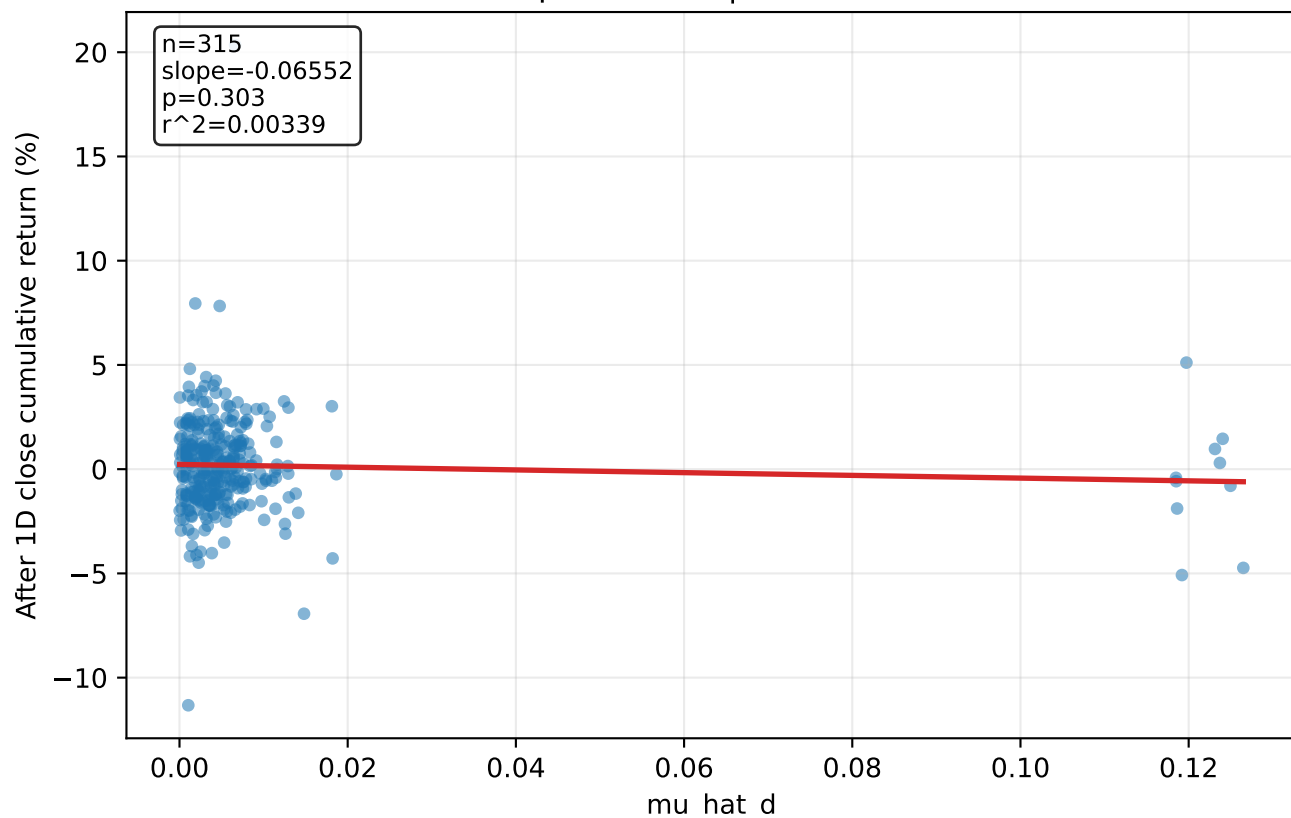
Zoomed future drift parameter mean path



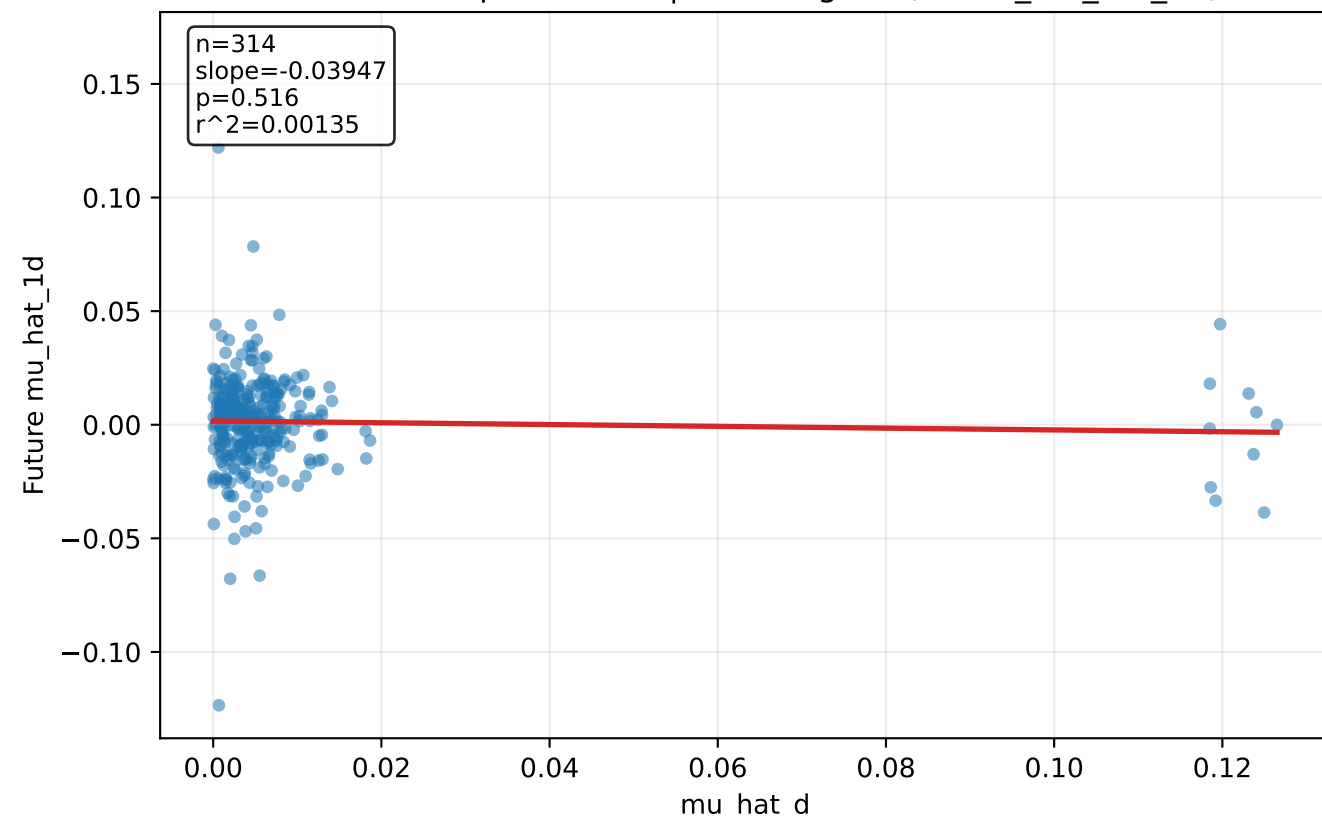
Zoomed future diffusion path as seq volatility (%)



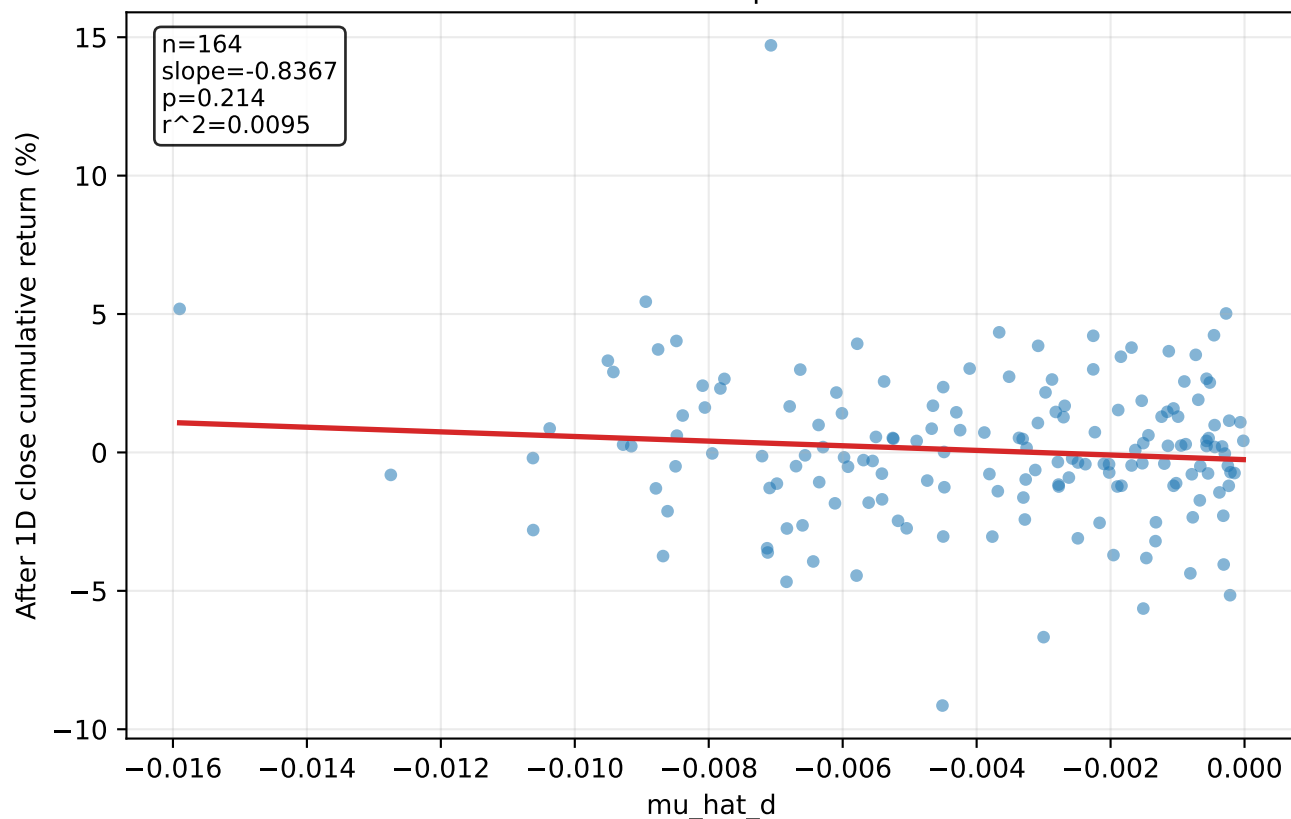
Detectable Upward Drift | After 1D Close Return



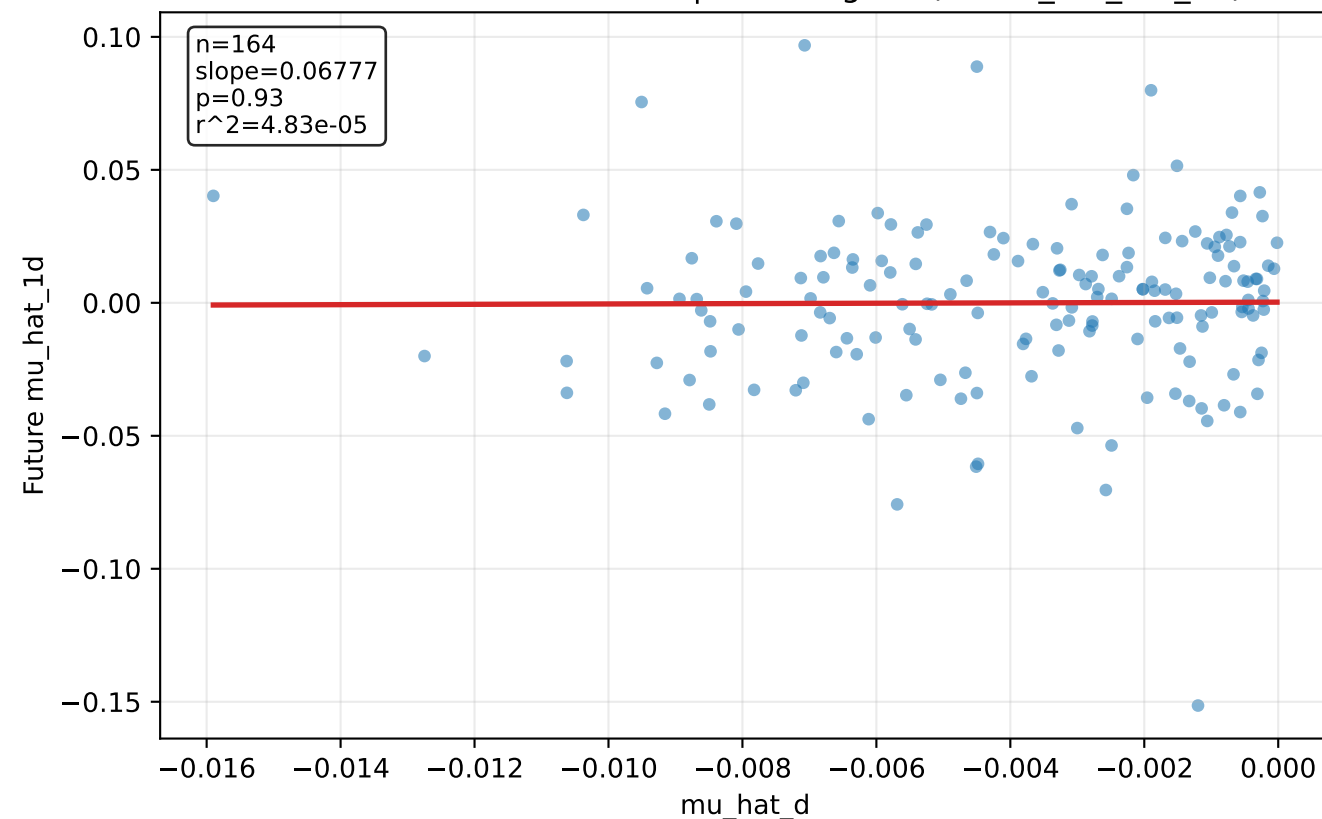
Detectable Upward Drift | Resulting Mu (future_mu_hat_1d)



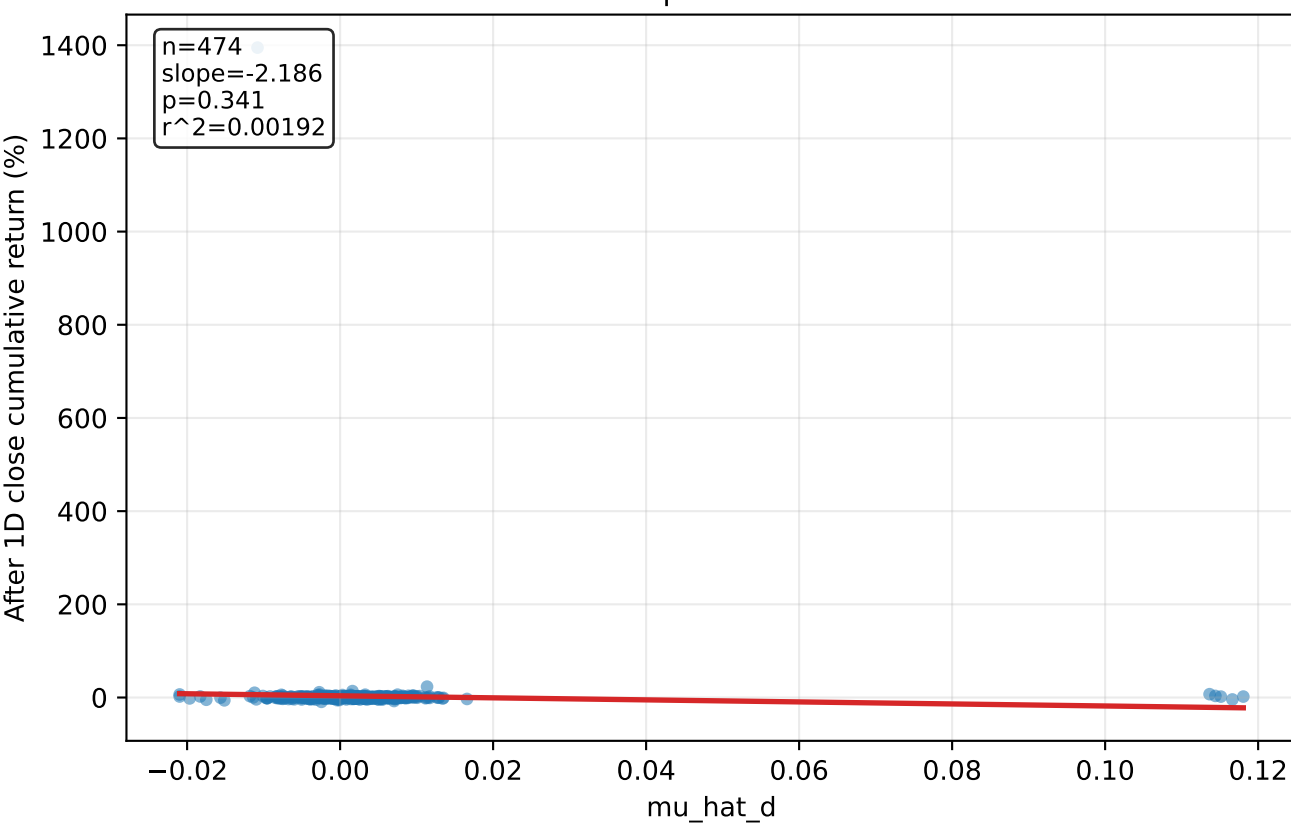
Detectable Downward Drift | After 1D Close Return



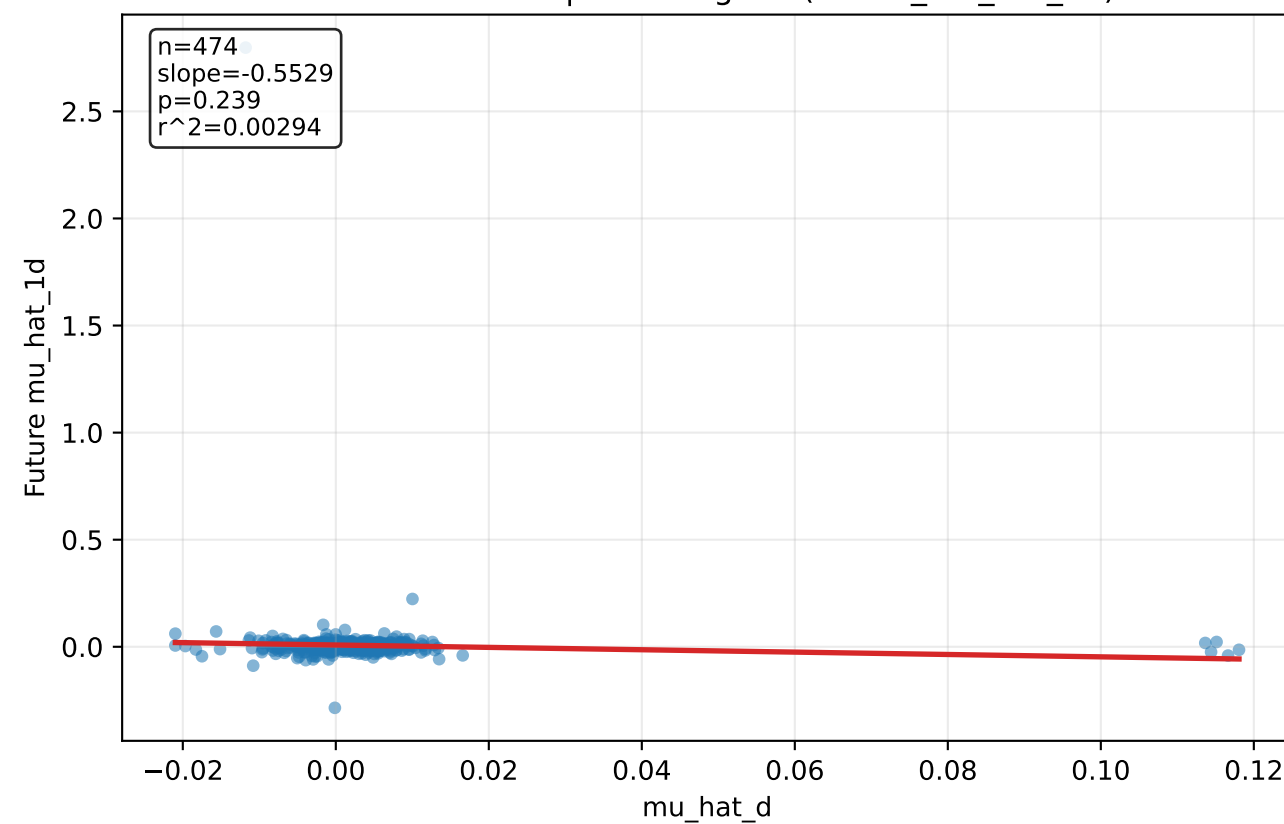
Detectable Downward Drift | Resulting Mu (future_mu_hat_1d)



Noise Dominant | After 1D Close Return



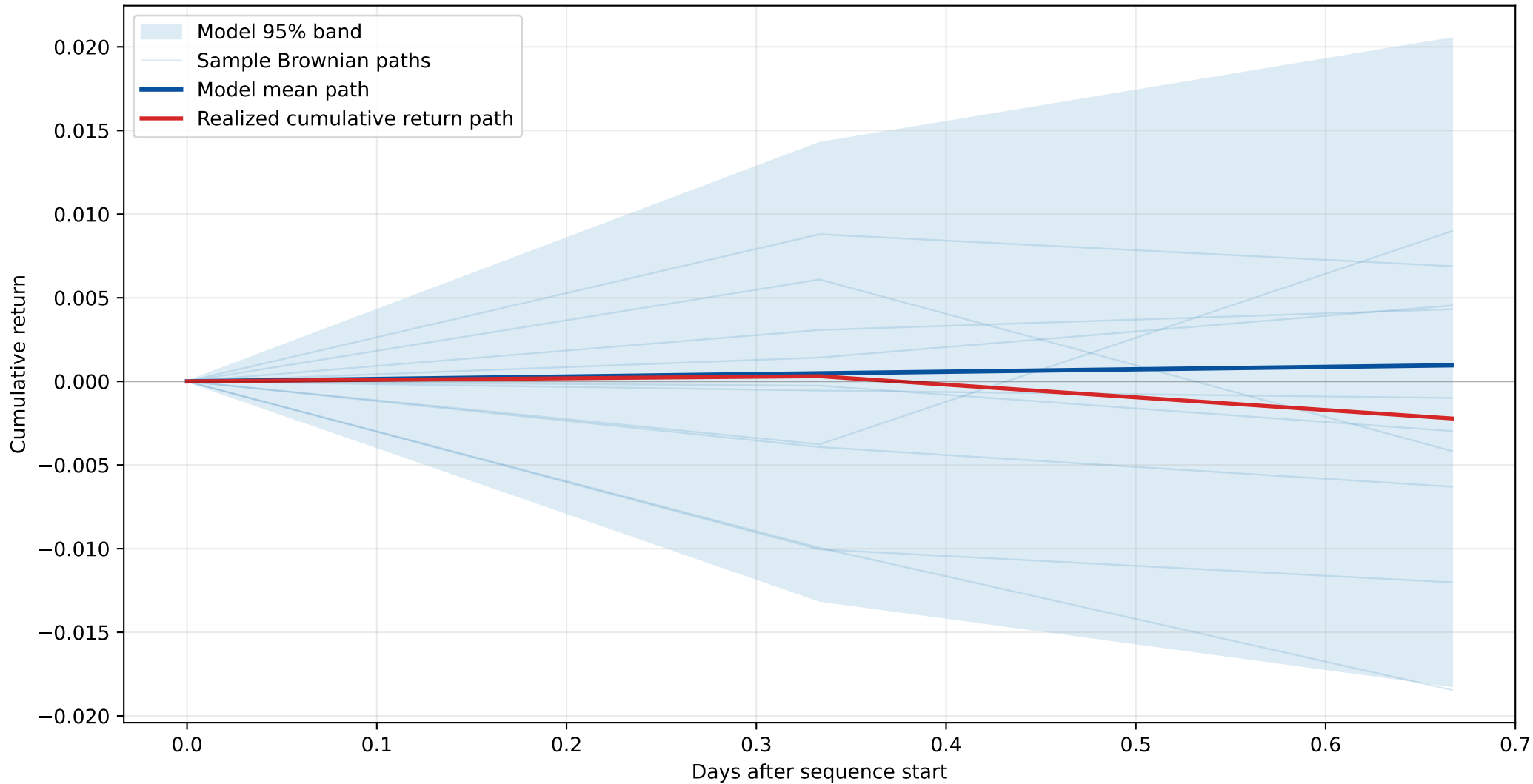
Noise Dominant | Resulting Mu (future_mu_hat_1d)



META Example Selection Rule

Examples are ranked by out-of-sample RMSE between the model-implied cumulative return path and the realized path. Selection is automatic and transparent: choose the lowest-RMSE valid daily examples under the current daily example. Example plots show the model-implied cumulative return path, a 95% band, simulated sample paths, and the realized path. Example annotations also report expected cumulative return and realized excursion metrics in basis points of the current daily example. Parameter estimation is based on the first-order Ait-Sahalia transition-density approximation applied to daily log returns.

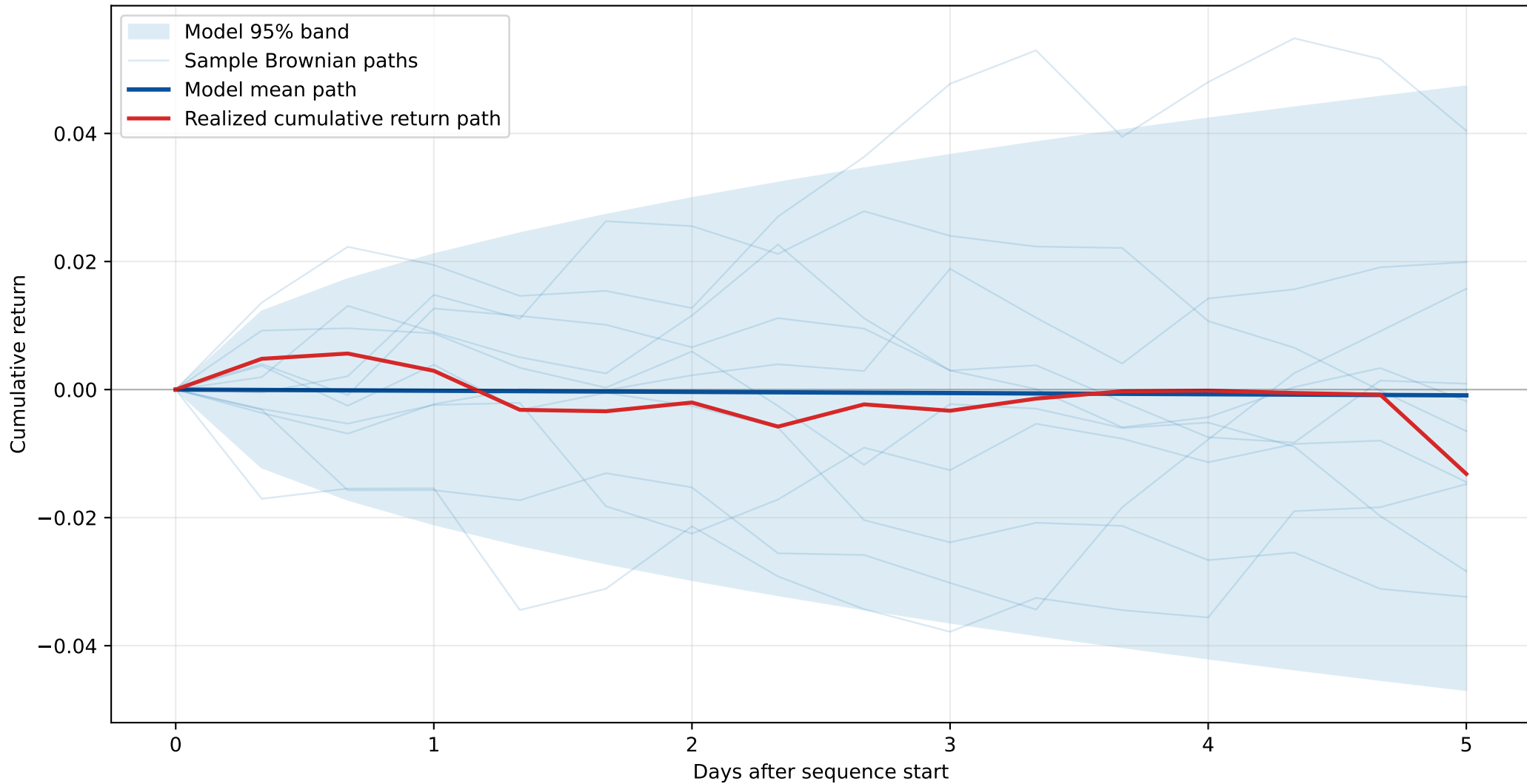
META | Past signal and future seq path



META Example Notes

signal_group = Signal group: detectable upward drift with general baseline and no abnormal volume
timestamp = 2025-12-30 15:59:00
freq = daily
mu_hat = 0.00144208
sigma_hat = 0.0120878
expected_cum_bps = 72.36
realized_cum_bps = -68.70
realized_max_up_bps = 0.00
realized_max_down_bps = -68.70
realized_max_abs_bps = 68.70
drift_sign = detectable_upward_drift
volume_level = low_volume
imbalance_side = neutral
exclusive_location = none
both_above_below_high = not_both_high

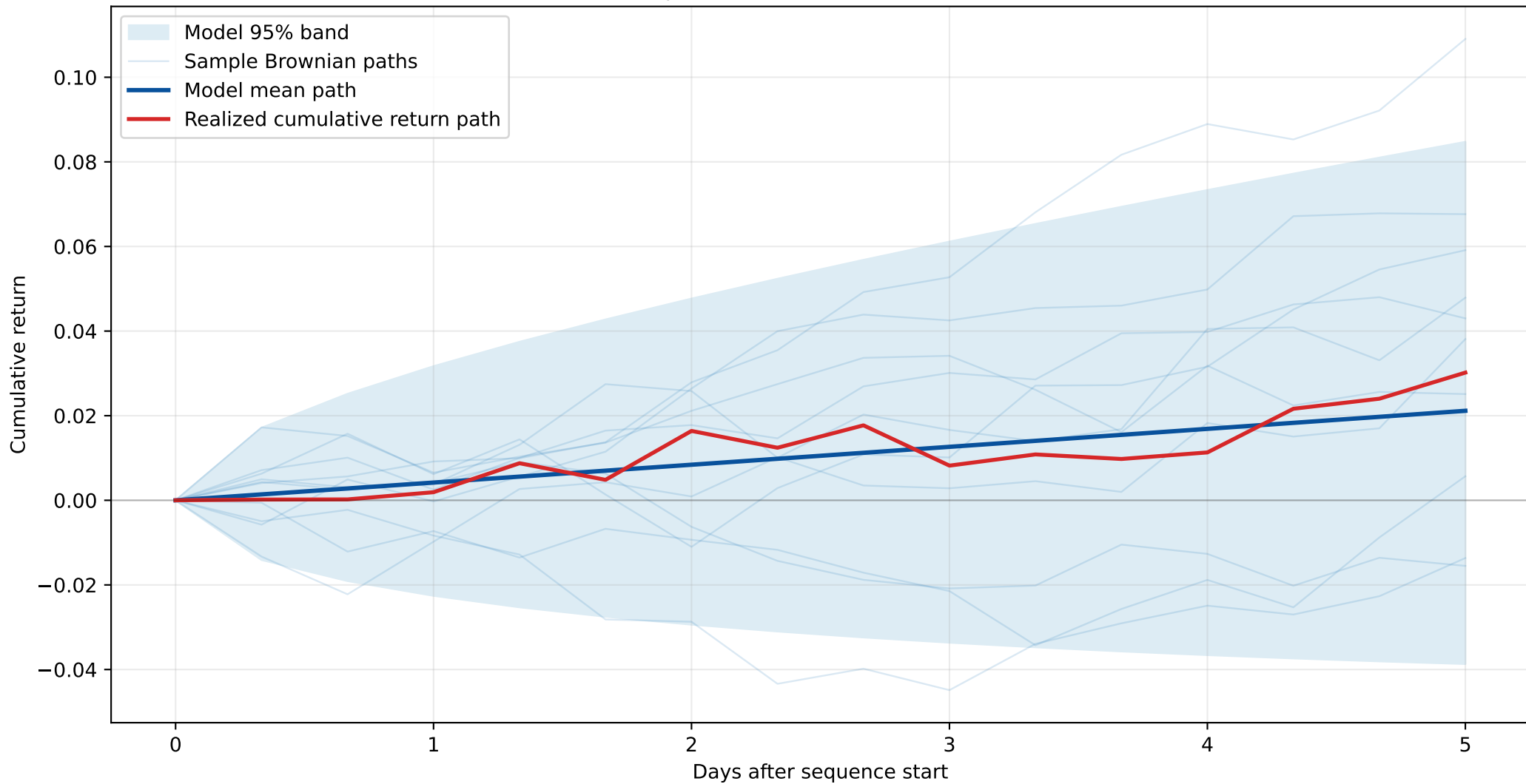
META | Past signal and future seq path



META Example Notes

signal_group = Signal group: noise dominant with general baseline and no abnormal volume
timestamp = 2021-08-06 15:59:00
freq = daily
mu_hat = -0.000180855
sigma_hat = 0.0107597
expected_cum_bps = -9.04
realized_cum_bps = 37.55
realized_max_up_bps = 63.78
realized_max_down_bps = 0.00
realized_max_abs_bps = 63.78
drift_sign = no_detectable_drift
volume_level = low_volume
imbalance_side = neutral
exclusive_location = none
both_above_below_high = not_both_high

META | Past signal and future seq path



META Example Notes

signal_group = Signal group: noise dominant with general baseline and no abnormal volume

timestamp = 2024-09-26 15:59:00

freq = daily

mu_hat = 0.0041868

sigma_hat = 0.013777

expected_cum_bps = 211.55

realized_cum_bps = 161.75

realized_max_up_bps = 161.75

realized_max_down_bps = -23.31

realized_max_abs_bps = 161.75

drift_sign = no_detectable_drift

volume_level = low_volume

imbalance_side = neutral

exclusive_location = none

both_above_below_high = not_both_high

META Interpretation

Local no-jump fit quality is evaluated using expected-path versus realized-path RMSE on future windows under

Largest cross-group realized-return separation appears in: daily.

Largest cross-group volatility separation appears in: daily.

Across reported group-difference tests, the 5% significance rate is 4.3%.

The daily detectable-drift trigger rate under the RSD screen is 54.5%.

Triggered-day regression links the daily μ_{hat_d} screen-passing sample to the existing after-1-day future

Economically noticeable effects should be judged by the scale of future μ_{hat} and σ_{hat} spreads, not c

Because this is a no-jump local diffusion benchmark on log VWAP with first-order approximate transition den